



Risk-averse two-stage stochastic programming with an application to disaster management

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ABSTRACT

Traditional two-stage stochastic programming is risk-neutral; that is, it considers the expectation as the preference criterion while comparing the random variables (e.g., total cost) to identify the best decisions. However, in the presence of variability risk measures should be incorporated into decision making problems in order to model its effects. In this study, we consider a risk-averse two-stage stochastic programming model, where we specify the conditional-value-at-risk (CVaR) as the risk measure. We construct two decomposition algorithms based on the generic Benders-decomposition approach to solve such problems. Both single-cut and multicut versions of the proposed decomposition algorithms are presented. We adapt the concepts of the value of perfect information (VPI) and the value of the stochastic solution (VSS) for the proposed risk-averse two-stage stochastic programming framework and define two stochastic measures on the VPI and VSS. We apply the proposed model to disaster management, which is one of the research fields that can significantly benefit from risk-averse two-stage stochastic programming models. In particular, we consider the problem of determining the response facility locations and the inventory levels of the relief supplies at each facility in the presence of uncertainty in demand and the damage level of the disaster network. We present numerical results to discuss how incorporating a risk measure affects the optimal solutions and demonstrate the computational effectiveness of the proposed methods.

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1. Introduction

Traditional two-stage stochastic programming considers the expectation as the preference criterion while comparing the random variables to find the best decisions; hence, it is a risk neutral approach. Two-stage stochastic programming with the expected recourse function has been applied in a wide range of applications. For different applications, we refer to Prékopa [1] and Birge and Louveaux [2], and references therein. We can justify the optimization of the expected value by the Law of Large Numbers for situations, where the same decisions under similar conditions are made repeatedly. Thus, the typical objective of minimizing the expected total cost may yield solutions that are good in the long run, but such results may perform poorly under certain realizations of the random data. Therefore, for non-repetitive decision making problems under uncertainty a risk-averse approach that considers the effects of the variability of random outcomes, such as the random total cost, would provide more robust solutions compared to a risk-neutral approach. Decisions such as location planning and disaster response management constitute examples of non-repetitive decisions.

The idea of introducing risk measures such as CVaR into the stochastic programming models is not new. However, incorporating risk measures into the objective functions in the framework of two-stage stochastic programming is a fairly recent research topic investigated among others by Ahmed [3,4], Schultz and Tiedemann [5], Fábán [6], Miller and Ruszczyński [7]. Ahmed [3,4] introduce the two-stage stochastic programming with mean-risk objectives. Ahmed [4] investigates several risk measures for which the mean-risk function preserves the convexity and thus, leads to computationally tractable methods. The author focuses on two classes of risk measures; the first class involves the absolute semideviation risk measure and the second one involves the quantile deviation risk measure, which is related to CVaR. Slight variants of the existing decomposition methods are applicable to solve the proposed problems, because the models do not involve any integer variables. Schultz and Tiedemann [5] focus on two-stage mixed-integer stochastic programming involving CVaR. In their model, the integer variables appear in the second-stage problem, and therefore, straightforward decomposition methods cannot be applied due to the non-convexity of the problem. They consider the split-variable formulation and develop a solution algorithm based on the Lagrangian relaxation of non-anticipativity. Recently, Miller and Ruszczyński [7] introduce a new type of risk-averse two-stage stochastic linear

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programming model, where the uncertainty is still present after the second-stage decisions are made. Due to the proposed extended structure, the authors consider compositions of conditional risk measures and develop variants of the existing decomposition methods, one of which is based on the multicut approach. We also refer to Fábíán [6] for different types of two-stage stochastic programming models involving CVaR. Fábíán [6] studies models where the unresolved uncertainty remains after the second-stage decisions are made, and the author proposes decomposition methods based on the polyhedral representation of CVaR used in Künzi-Bay and Mayer [8]. We note that Künzi-Bay and Mayer [8] reformulate the single-stage CVaR minimization problem as a two-stage stochastic programming and propose a new algorithm, which is a variant of the L-shaped method and it is based on the optimality cuts proposed by Klein Haneveld and Van der Vlerk [9] for integrated chance constraints. This study is not directly related to ours, because Künzi-Bay and Mayer [8] consider a single-stage stochastic programming problem involving CVaR.

In this paper, we consider the risk-averse two-stage stochastic programming framework similar to those of Ahmed [4] and Schultz and Tiedemann [5]. We characterize the inherent randomness by a finite set of scenarios and specify CVaR as the risk measure in the two-stage mean-risk analysis. Different than Schultz and Tiedemann [5] we do not consider any integer variables in the second-stage problem and therefore, the standard decomposition methods apply in our study as in Ahmed [4]. The *L-shaped method* proposed by Van Slyke and Wets [10] is a widely applied Benders-decomposition approach [11] to solve the two-stage stochastic programming problems with the expected recourse functions for the case of a finite probability space. For a detailed discussion, we refer to Van Slyke and Wets [10], Prékopa [1] and Birge and Louveaux [2]. We reformulate the two-stage stochastic programming problem with the CVaR measure on the total cost as a large scale linear programming problem and in order to solve it we develop an algorithm, which is a variant of the L-shaped decomposition method. As an alternate solution algorithm, the generic algorithm proposed by Ahmed [4] is adapted to CVaR. Such decomposition algorithms allow us to handle a large number of scenarios, which is crucial in obtaining a good representation of the randomness. We extend both proposed decomposition algorithms by replacing the single-cuts by their multicut versions.

Stochastic programming approach to model decision making problems in the presence of uncertainty is generally known to be computationally challenging. Therefore, some decision makers may prefer to solve simpler deterministic versions of the problems instead of a stochastic programming problem. For example, such a decision maker can solve the deterministic problem where all the random parameters are assumed to take their expected values. A natural question is whether solving such simpler versions of the problem provides reasonably good solutions. There exist two well-known measures, the expected value of perfect information (EVPI) and the VSS, which are designed to indicate whether it is worth to model the randomness using the risk-neutral two-stage stochastic programming approach. For details on these stochastic measures we refer the reader to Birge and Louveaux [2]. We adapt these concepts for the proposed two-stage mean-risk stochastic programming model and define new measures on the VPI and VSS.

Risk-averse two-stage stochastic programming with mean-risk functions has a significant potential to be applied in different fields. Since it is a fairly recent development, it has only been applied to just a few problems. For instance, Schultz and Tiedemann [5], Schultz and Neise [12], and Liu et al. [13] study problems in chemical engineering, energy optimization, and

transportation network protection, respectively. In this study, we apply the proposed framework to disaster management that is one of the research fields that can significantly benefit from the risk-averse two-stage models. Several two-stage stochastic programming problems are proposed for disaster management among others by Barbarosoğlu and Arda [14], Balcik and Beamon [15], and Rawls and Turnquist [16]. However, all such existing models focus on the expected values and considering only the expected values may not be good enough for rarely occurring disaster events. To the best of our knowledge, developing a risk-averse two-stage stochastic model in the disaster management literature is novel. In particular, we consider the problem of determining the response (storage) facility locations and the inventory levels of the relief supplies at each facility in the presence of uncertainty in demand and the damage level of the disaster network. We assume that when a disaster occurs the roads and facilities may be damaged and therefore, the transportation capacities and the available amount of preallocated supplies may vary according to the severity of the disaster. We focus on an existing two-stage stochastic programming formulation of a disaster location-allocation model that has been recently proposed by Rawls and Turnquist [16] and we extend it by incorporating the CVaR as the risk measure on the total cost. Our proposed model is a significant and non-trivial extension, because it incorporates a risk measure in addition to the expectation criterion. It has been quite popular in different fields to develop models involving CVaR; for instances, we refer to Rockafellar and Uryasev [17], Chen et al. [18], and Schultz and Tiedemann [5]. Solving the proposed problem for different risk parameters provides alternate location and allocation decisions and decision makers can evaluate those decisions with respect to costs, the quality of service and also their risk preferences. We conduct an extensive computational analysis for the case study constructed in Rawls and Turnquist [16] to illustrate how the risk parameters affect the location and allocation policies and show that a disaster management model involving a risk measure can still be solved efficiently for reasonably large problem instances.

The contributions of this study are (i) developing decomposition algorithms that are specific for CVaR in the two-stage mean-risk stochastic programming, (ii) adapting the concepts of the VPI and the VSS for the two-stage mean-risk stochastic programming models, (iii) introducing a risk-averse two-stage disaster preparedness model, (iv) showing that resulting risk-averse disaster management model can still be solved efficiently for reasonably large problem instances. The proposed general framework can be applied to other problems, and this approach would provide the decision makers with risk-averse decisions that would perform better in the presence of variability compared to a risk-neutral approach. The paper is organized as follows. In Section 2, we discuss how to model risk in the two-stage stochastic programming framework in general. In Section 3, we first develop single-cut decomposition algorithms for the proposed model and then describe their multicut versions. Then in Section 4, we discuss the concepts of the EVPI and the VSS and define similar stochastic measures for the two-stage mean-risk stochastic programming models. In Section 5, we describe the existing disaster location-allocation model under the expected total cost criterion and then introduce our risk-averse model. In Section 6, we present computational results for the case study given in Rawls and Turnquist [16] to illustrate how the location and allocation solutions change with respect to different risk preferences. In addition, numerical results for a number of problem instances are provided to demonstrate the computational effectiveness of the proposed solution methods. We conclude the paper in Section 7.

2. Two-stage mean-risk stochastic programming framework

An abstract probability space is denoted by $(\Omega, \mathcal{F}, P)$, where Ω is the sample space, $\mathcal{F}$ is a σ -algebra on Ω and P is a probability measure on Ω . We consider the case of a finite probability space, where $\Omega = \{\omega^1, \dots, \omega^N\}$ with corresponding probabilities $p^1, \dots, p^N$.

The general form of the risk-neutral two-stage stochastic linear programming problem reads

$$\min_{\mathbf{x} \in X} \mathbb{E}(f(\mathbf{x}, \omega)) = \min_{\mathbf{x} \in X} \mathbf{c}^T \mathbf{x} + \mathbb{E}(Q(\mathbf{x}, \xi(\omega))), \quad (1)$$

where $f(\mathbf{x}, \omega) = \mathbf{c}^T \mathbf{x} + Q(\mathbf{x}, \xi(\omega))$ is the total cost function of the first-stage problem and

$$Q(\mathbf{x}, \xi^s) = \min_{\mathbf{y}^s} \{(\mathbf{q}^s)^T \mathbf{y}^s : \mathbf{T}^s \mathbf{x} + \mathbf{W}^s \mathbf{y}^s = \mathbf{h}^s, \mathbf{y}^s \geq 0\} \quad (2)$$

is the second-stage problem corresponding to the realization of the random data $\xi(\omega)$ for the elementary event ω^s , denoted by $\xi^s = (\mathbf{q}^s, \mathbf{T}^s, \mathbf{W}^s, \mathbf{h}^s)$. Here $\mathbf{x}$ and $\mathbf{y}$ are the vectors of first-stage and second-stage decision variables, respectively. Suppose that all the matrices have the appropriate dimensions. In (1) and (2) we consider the linear equalities and objective functions. Note that the linear programming structure of the second-stage problem is crucial to develop efficient cutting plane algorithms. We assume that $X \subset \mathbb{R}_+^n$ is a non-empty set of feasible decisions, $Q(\mathbf{x}, \xi(\omega)) > -\infty$ for all $\omega \in \Omega$, and the second-stage problem (2) may be infeasible for some first-stage decision $\mathbf{x} \in X$. Observe that the first-stage decisions are deterministic and the second-stage decisions are allowed to depend on the elementary events, i.e., $\mathbf{y}^s = \mathbf{y}(\omega^s)$, $s = 1, \dots, N$. Basically, the second-stage decisions represent the operational decisions, which change depending on the realized values of the random data. The objective function $Q(\mathbf{x}, \xi(\omega))$ of the second-stage problem (2), also known as the recourse (cost) function, is a random variable and therefore, the total cost function $f(\mathbf{x}, \omega)$ is a random variable and determining the optimal decision vector $\mathbf{x}$ leads to the problem of comparing random cost variables $\{f(\mathbf{x}, \omega)\}_{\mathbf{x} \in X}$. Comparing random variables is one of the main interests of decision theory in the presence of uncertainty. Since we focus on the total cost, smaller values of $f(\mathbf{x}, \omega)$ are preferred. While comparing random variables it is crucial to consider the effect of variability and specify the preference relations among random variables using risk measures. One of the main approaches in the practice of decision making under risk uses mean-risk models. In these models one minimizes the mean-risk function, which involves a specified risk measure $\rho : \mathcal{Z} \rightarrow \mathbb{R}$, where ρ is a functional and $\mathcal{Z}$ is a linear space of $\mathcal{F}$ -measurable functions on the probability space $(\Omega, \mathcal{F}, P)$:

$$\min_{\mathbf{x} \in X} \{\mathbb{E}(f(\mathbf{x}, \omega)) + \lambda \rho(f(\mathbf{x}, \omega))\}.$$

In this approach, λ is a non-negative trade-off coefficient representing the exchange rate of mean cost for risk. We also refer to it as the risk coefficient, which is specified by decision makers according to their risk preferences. The well-known model by Markowitz [19] uses the variance as the risk measure. One of the problems associated with this mean-variance formulation is that it treats under-and-over-performance equally. Usually, when typical dispersion statistics such as variance are used as risk measures, the mean-risk approach may lead to inferior solutions (see Remark 2.1). In order to remedy this drawback, models with alternative asymmetric risk measures such as downside risk measures have been proposed (see e.g., [20]). Among the well-known downside risk measures we focus on a recently popular and widely applied risk measure called conditional value-at-risk. For alternate risk measures that lead to computationally tractable two-stage mean-risk models, we refer to Ahmed [4].

In our study, we say that the decision vector $\mathbf{x}$ is efficient (in the mean-risk sense) if and only if for a given level of expected cost, $f(\mathbf{x}, \omega)$ has the lowest possible CVaR, and for a given level of CVaR it has the lowest possible expected cost. One can construct the mean-risk efficient frontier by finding the efficient solutions for different risk coefficients. Thus, we consider the following two-stage mean-risk stochastic programming problem:

$$\min_{\mathbf{x} \in X} \{\mathbb{E}(f(\mathbf{x}, \omega)) + \lambda \text{CVaR}_\alpha(f(\mathbf{x}, \omega))\}, \quad (3)$$

where CVaR_α denotes the conditional value-at-risk at level α . Next, we present the relevant definitions.

Definition 2.1. Let $F_Z(\cdot)$ represent the cumulative distribution function of a random variable Z . In the financial literature, the α -quantile

$$\inf\{\eta \in \mathbb{R} : F_Z(\eta) \geq \alpha\}$$

is called the value-at-risk (VaR) at the confidence level α and denoted by $\text{VaR}_\alpha(Z)$, $\alpha \in (0, 1]$.

The conditional value-at-risk, also called mean excess loss or tail VaR, at level α is, in a simple way, defined as follows [21]:

$$\text{CVaR}_\alpha(Z) = \mathbb{E}(Z | Z \geq \text{VaR}_\alpha(Z)). \quad (4)$$

This definition provides a clear understanding of the concept of CVaR: $\text{CVaR}_\alpha(Z)$ is the conditional expected value exceeding the value-at-risk at the confidence level α . In the cost minimization context, VaR_α is the α -quantile of the distribution of the cost and it provides an upper bound that is exceeded only with a small probability of $1-\alpha$. On the other hand, $\text{CVaR}_\alpha(Z)$ is a measure of severity of the cost if it is more than $\text{VaR}_\alpha(Z)$. The formula (4) is precise if $\text{VaR}_\alpha(Z)$ is not an atom of the distribution of Z . A more precise description is given in the next definition [17,21].

Definition 2.2. The conditional-value-at-risk of a random variable Z at the confidence level α is given by

$$\text{CVaR}_\alpha(Z) = \inf_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1-\alpha} \mathbb{E}([Z-\eta]_+) \right\}, \quad (5)$$

where we let $[z]_+ = \max\{0, z\}$, $z \in \mathbb{R}$. It is well-known that the infimum in (5) is attained at a α -quantile of Z (see, e.g., [20]). Our first proposed algorithm utilizes this result while calculating the CVaR value for the realized values of the recourse function, $Q(\mathbf{x}, \xi^s)$, $s = 1, \dots, N$, associated with a candidate first-stage decision vector $\mathbf{x}$.

Proposition 2.1. For the case of a finite probability space, where $\Omega = \{\omega^1, \dots, \omega^N\}$ with corresponding probabilities $p^1, \dots, p^N$, we can equivalently reformulate the mean-risk problem

$$\min_{\mathbf{x} \in X} \{\mathbb{E}(f(\mathbf{x}, \omega)) + \lambda \text{CVaR}_\alpha(f(\mathbf{x}, \omega))\}$$

as the following linear programming problem:

$$\begin{aligned} \min \quad & (1+\lambda)\mathbf{c}^T \mathbf{x} + \sum_{s=1}^N p^s (\mathbf{q}^s)^T \mathbf{y}^s + \lambda \left(\eta + \frac{1}{1-\alpha} \sum_{s=1}^N p^s v^s \right) \\ \text{subject to} \quad & \mathbf{W}^s \mathbf{y}^s = \mathbf{h}^s - \mathbf{T}^s \mathbf{x}, \quad s = 1, \dots, N, \\ & \mathbf{x} \in X, \\ & \mathbf{y}^s \geq 0, \quad s = 1, \dots, N, \\ & v^s \geq (\mathbf{q}^s)^T \mathbf{y}^s - \eta, \quad s = 1, \dots, N, \\ & \eta \in \mathbb{R}, v^s \geq 0, \quad s = 1, \dots, N. \end{aligned}$$

Proof. It is easy to show that (see, eg., [22,23]) CVaR_α is translation invariant, i.e.,

$$\text{CVaR}_\alpha(Z+a) = \text{CVaR}_\alpha(Z) + a$$

for $a \in \mathbb{R}$ and $Z \in \mathcal{Z}$. Then, we have

$$\begin{aligned} \text{CVaR}_\alpha(f(\mathbf{x}, \omega)) &= \mathbf{c}^T \mathbf{x} + \text{CVaR}_\alpha(Q(\mathbf{x}, \xi(\omega))) \\ \text{and} \\ \mathbb{E}(f(\mathbf{x}, \omega)) + \lambda \text{CVaR}_\alpha(f(\mathbf{x}, \omega)) \\ &= \mathbf{c}^T \mathbf{x} + \mathbb{E}(Q(\mathbf{x}, \xi(\omega))) + \lambda(\mathbf{c}^T \mathbf{x} + \text{CVaR}_\alpha(Q(\mathbf{x}, \xi(\omega)))) \\ &= (1 + \lambda)\mathbf{c}^T \mathbf{x} + \mathbb{E}(Q(\mathbf{x}, \xi(\omega))) + \lambda \text{CVaR}_\alpha(Q(\mathbf{x}, \xi(\omega))). \end{aligned} \quad (6)$$

For the given finitely many realizations of the random data $\xi(\omega)$, we have the realizations of the recourse function as $Q(\mathbf{x}, \xi^1) = (q^1)^T \mathbf{y}^1, \dots, Q(\mathbf{x}, \xi^N) = (q^N)^T \mathbf{y}^N$. Then by the description of the second-stage problem (2) and by Definition (2.2), the assertion follows. $\square$

Note that we can interpret the variable η as a first-stage variable and the excess variables, $v^s, s = 1, \dots, N$, as second-stage variables. For a finite probability space, reformulating a risk-neutral linear two-stage stochastic programming problem without integer restrictions as a large-scale linear programming problem is well-known as discussed in depth by Prékopa [1] and Birge and Louveaux [2]. Proposition 2.1 is significant to show that the two-stage stochastic programming problem involving CVaR can also be formulated as a linear programming problem. A similar formulation is also presented in Schultz and Tiedemann [5].

Remark 2.1 (Consistency with the second order stochastic dominance (SSD)). The stochastic dominance relations are fundamental and widely applied concepts in comparing random variables. For a review on stochastic dominance rules and other types of measures to compare random variables, we refer the reader to Shaked and Shanthikumar [24] and Müller and Stoyan [25] and the references therein. We say that the mean-risk model $(E(Z), \rho(Z))$ is consistent with SSD if the following relation holds:

$$Z_1 \text{ dominates } Z_2 \text{ in the second order} \Rightarrow E(Z_1) \leq E(Z_2) \quad \text{and} \quad \rho(Z_1) \leq \rho(Z_2).$$

The mean-risk model is in general not consistent with stochastic dominance rules. For example, when the variance is used as a risk measure the mean-risk model is not consistent with SSD. Ogryczak and Ruszczyński [20,26] have studied the consistency of mean-risk models with SSD relation and they have proved that the mean-risk model with CVaR is consistent with SSD. This result emphasizes the importance of the mean-risk models involving CVaR as a risk measure.

It is easy to show that $Q(\mathbf{x}, \xi(\omega))$ is convex in $\mathbf{x}$ for all $\omega \in \Omega$, and so is $f(\mathbf{x}, \omega)$. Moreover, for a given $\alpha \in (0, 1)$ the mean-risk function in (3) is convexity preserving for all $\lambda \geq 0$ (see [4]). For the second-stage problems the duality theory of linear programming can be used to obtain outer approximations of the mean-risk function of the recourse cost in (6). Constructing the outer approximations of the recourse function is basically the one of the main ideas of the *L-shaped method* proposed by Van Slyke and Wets [10]. In this study, we develop two variants of the L-shaped method to solve the two-stage stochastic programming problem (3) involving the CVaR measure on the random total cost.

3. Decomposition algorithms

In this section, we first present two decomposition algorithms to solve problem (3). Both algorithms are variants of the L-shaped method; the optimality (objective) cuts are generated to approximate the optimal mean-risk function of the recourse cost, instead of just the expected recourse cost. Initially, we describe the algorithms based on the single-cut method, which aggregates the dual information of the second-stage problem over all the realizations of the

random data to generate optimality cuts. In the first algorithm, we introduce additional variables to construct the optimality cuts, while in the second algorithm we avoid introducing new variables by using the subgradients of the recourse function. We extend both of the proposed decomposition algorithms by replacing the single-cuts by their multicut versions. The multicut approach disaggregates the optimality cuts to avoid the loss of the dual information associated with each realization of the second-stage problem. Since CVaR is only involved in the objective function and second-stage decision variables are all continuous, the feasibility cuts used for the expectation case directly apply here as well.

We would like to emphasize that the proposed decomposition algorithms can directly be used to solve the two-stage mean-risk stochastic programming problems, where there also exist integer variables in the first-stage problem (1). The only difference is that at each iteration of the decomposition algorithm one needs to solve a mixed-integer programming master problem instead of a linear programming master problem. This may pose a computational challenge, since solving the master problem several times can be time consuming for large problem instances. To remedy this issue, one may incorporate a heuristic into the decomposition algorithms to obtain a near-optimal solution to the master problem more quickly. Alternatively, one may utilize a branch-and-cut scheme in the master problem. Such a method, referred to as the “integer L-shaped method”, has been developed by Laporte and Louveaux [27] to solve stochastic integer problems with binary first stage variables. However, in this study we restrict our attention to the structure of the optimality cuts and design decomposition methods to deal with a large number of scenarios. For a given specific application, our framework can be customized by incorporating features that improve the computational efficiency of the master problem.

3.1. Basic decomposition algorithm for CVaR

In this algorithm, we generate the t th optimality cut associated with CVaR of the recourse cost by introducing new variables η^t and $v^{st}, s = 1, \dots, N$. At an iteration of the algorithm, if the second-stage problems for the current first-stage solution and all the realizations of $\xi(\omega)$ are feasible, we solve all the second-stage problems to optimality. Suppose that at the current iteration, we construct the t th optimality cut and $\mathbf{u}^{st}$ is the dual vector associated with the optimal basis of the second-stage problem for realization ξ^s . Then, $(\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x})$ is the optimal recourse function value for the realization ξ^s and the CVaR value of the recourse cost, is given by

$$\text{CVaR}_\alpha(Q(\mathbf{x}, \xi(\omega))) = \inf_{\eta^t \in \mathbb{R}} \left\{ \eta^t + \frac{1}{1-\alpha} \sum_{s=1}^N p^s [(\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}) - \eta^t]_+ \right\}.$$

In the master problem of the decomposition algorithm, the new variable η^t and the new variables $v^{st}, s = 1, \dots, N$, are introduced to represent the α -quantile of the recourse cost and $[(\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}) - \eta^t]_+, s = 1, \dots, N$, respectively. The optimality cuts are constructed by aggregating the dual information of the second-stage problem over all the realizations of the random data. By considering the cuts associated with the first-stage solutions obtained so far, we iteratively improve the lower bound on the optimal mean-risk function of the total cost. Thus, the master problem of this decomposition algorithm becomes

$$\min \quad (1 + \lambda)\mathbf{c}^T \mathbf{x} + \theta_1 + \lambda \theta_2 \quad (7)$$

$$\text{subject to} \quad \mathbf{x} \in X, \quad (8)$$

$$\text{optimality cuts} \quad \theta_1 \geq \sum_{s=1}^N p^s (\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}), \quad t = 1, \dots, \tau, \quad (9)$$

$$\text{optimality cuts } \theta_2 \geq \eta^t + \frac{1}{1-\alpha} \sum_{s=1}^N p^s v^{st}, \quad t=1, \dots, \tau, \quad (10)$$

$$v^{st} \geq (\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}) - \eta^t, \quad s=1, \dots, N, t=1, \dots, \tau, \quad (11)$$

$$v^{st} \geq 0, \quad s=1, \dots, N, t=1, \dots, \tau, \quad (12)$$

$$\eta^t \in \mathbb{R}, \quad t=1, \dots, \tau, \quad (13)$$

$$\text{feasibility cuts } (T^s)^T \sigma^{g_s} \mathbf{x} - (\mathbf{h}^s)^T \sigma^{g_s} \geq 0, \quad s=1, \dots, N, g_s=1, \dots, G_s. \quad (14)$$

Here τ and G_s denote the number of optimality and feasibility cuts constructed up to the current iteration, respectively. σ^{g_s} denotes the dual vector associated with the optimal basis of the feasibility problem of the second-stage problem for realization ξ^s and it is used to generate the corresponding g_s th feasibility cut. For the details of the feasibility problem and the feasibility cuts the reader is referred to Prékopa [1] and Birge and Louveaux [2]. Notice that we solve the master problem $\sum_{s=1}^N G_s + \tau$ times to obtain the formulation above. A two-stage stochastic programming problem is said to have complete recourse when its second-stage problem is feasible for any given first-stage decision vector. For such problems, we do not need to consider the feasibility cuts.

Algorithm 1. Decomposition Algorithm 1.

- 1: Initialize $\tau = 0$, $G_s = 0$, $s = 1, \dots, N$.
- 2: Solve the master problem (7)–(14). Let $(\bar{\mathbf{x}}, \bar{\theta}_1, \bar{\theta}_2)$ be the optimal solution and $\bar{\theta} = \bar{\theta}_1 + \lambda \bar{\theta}_2$. (When $\tau = 0$ ignore θ_1 , θ_2 , and the optimality cuts. When $G_s = 0$ ignore the feasibility cuts.)
- 3: (If the problem has complete recourse skip this step.)

Check whether the second-stage problems are all feasible for $\bar{\mathbf{x}}$ by solving the following feasibility problem for realization ξ^s , $s = 1, \dots, N$:

$$\min_{(\mathbf{v}^+, \mathbf{v}^-)} \left\{ \sum_j (v_j^+ + v_j^-) : W^s \mathbf{y}^s + \mathbf{v}^+ - \mathbf{v}^- = \mathbf{h}^s - T^s \bar{\mathbf{x}}, \mathbf{y}^s \geq 0, \mathbf{v}^+ \geq 0, \mathbf{v}^- \geq 0 \right\}. \quad (15)$$

If the optimal objective function value of (15) is zero for all $s = 1, \dots, N$, then all the second-stage problems are feasible and go to Step 4. Otherwise pick an infeasible problem, say s th problem. Then, let $G_s = G_s + 1$ and introduce the feasibility cut for the dual vector σ^{G_s} associated with the optimal basis in problem (15).

Repeat Steps 2 and 3 until all the second-stage problems are feasible.

- 4: Solve all the second-stage problems:

$$\min_{\mathbf{y}^s} \{ (\mathbf{q}^s)^T \mathbf{y}^s : W^s \mathbf{y}^s = \mathbf{h}^s - T^s \bar{\mathbf{x}}, \mathbf{y}^s \geq 0 \}. \quad (16)$$

Let $\tau = \tau + 1$ and $\mathbf{u}^{s\tau}$ denote the dual vector corresponding to the optimal basis in problem (16).

- 5: Observe that $(\mathbf{u}^{s\tau})^T (\mathbf{h}^s - T^s \bar{\mathbf{x}})$, $s = 1, \dots, N$, are the realizations of $Q(\bar{\mathbf{x}}, \omega)$.

Find the α -quantile of $Q(\bar{\mathbf{x}}, \omega)$, denoted by $\bar{\eta}_{\alpha}$, and calculate $\text{CVaR}_{\alpha}(Q(\bar{\mathbf{x}}, \omega))$:

$$\text{CVaR}_{\alpha}(Q(\bar{\mathbf{x}}, \omega)) = \bar{\eta}_{\alpha} + \frac{1}{1-\alpha} \left(\sum_{s=1}^N p^s [Q(\bar{\mathbf{x}}, \omega^s) - \bar{\eta}_{\alpha}]_+ \right).$$

- 6: Calculate the mean-risk function value of the recourse cost at the current solution:

$$\theta^* = \sum_{s=1}^N p^s (\mathbf{u}^{s\tau})^T (\mathbf{h}^s - T^s \bar{\mathbf{x}}) + \lambda \text{CVaR}_{\alpha}(Q(\bar{\mathbf{x}}, \omega)).$$

- 7: if $\bar{\theta} \geq \theta^*$ then

8: Stop. $\bar{\mathbf{x}}$ is the optimal first-stage decision vector.

9: else

10: Introduce the optimality cuts for $\mathbf{u}^{s\tau}$ and go to Step 2.

11: end if

In the formulation of the master problem, we approximate the expected recourse function and the associated CVaR value separately. Thus, the aggregate dual information is used to generate two single-cuts; one for the expectation term and one for the CVaR term. Alternatively, we can approximate $\mathbb{E}(Q(\bar{\mathbf{x}}, \xi(\omega))) + \lambda \text{CVaR}_{\alpha}(Q(\bar{\mathbf{x}}, \xi(\omega)))$ by introducing $\hat{\theta} = \theta_1 + \lambda \theta_2$ and the combined optimality cuts

$$\hat{\theta} \geq \sum_{s=1}^N p^s (\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}) + \lambda \left(\eta^t + \frac{1}{1-\alpha} \sum_{s=1}^N p^s v^{st} \right), \quad t=1, \dots, \tau. \quad (17)$$

In this case, a single-cut is generated for the combination of the expectation and the CVaR terms. However, as we observed in our preliminary experiments, it is in general computationally more efficient to consider the separate optimality cuts given in (9) and (10). The description of this decomposition algorithm is presented in Algorithm 1. The correctness and finiteness of the algorithm can be proved similar to those of the traditional L-shaped algorithm; for details we refer to Van Slyke and Wets [10], Prékopa [1], and Birge and Louveaux [2].

Remark 3.1. We relax the stopping condition $\bar{\theta} \geq \theta^*$ (Step 7) and terminate the algorithm with a positive optimality gap. Let $(\bar{\mathbf{x}}, \bar{\theta})$ be the optimal solution and $\bar{f}$ be the optimal objective function value of the master problem at an iteration of the algorithm. The decision vector $\bar{\mathbf{x}}$ is a feasible solution of the original problem and the associated objective function value is equal to $\hat{f} = \bar{f} - \bar{\theta} + \theta^*$, where θ^* is the value of $\{\mathbb{E}(Q(\bar{\mathbf{x}}, \omega)) + \lambda \text{CVaR}_{\alpha}(Q(\bar{\mathbf{x}}, \omega))\}$. It is easy to see that $\hat{f}$ is the best objective function value found so far and $\bar{f}$ is the best known lower bound on the optimal objective function value. Then, an upper bound on the relative optimality can be estimated as follows:

$$\text{UBROG} = \frac{\hat{f} - \bar{f}}{\bar{f}} = \frac{\theta^* - \bar{\theta}}{\bar{f}}.$$

We stop the algorithm when $\text{UBROG} \leq \varepsilon$, where ε is a specified stopping tolerance value.

3.2. Subgradient-based decomposition algorithm

The first proposed algorithm introduces additional variables to represent the CVaR values, which may be computationally expensive if a very large number of scenarios is used to characterize the randomness. As an alternative we consider another algorithm, where we approximate the CVaR value of the recourse function by constructing the subgradient inequalities which replace the optimality cuts (10)–(13) used in Algorithm 1. This algorithm has been adapted from the general algorithm proposed by Ahmed [4] for the class of risk measures involving the quantile deviation risk measure.

Let $v(\mathbf{x}, \eta) = E([Q(\mathbf{x}, \xi(\omega)) - \eta]_+)$, where $\eta \in \mathbb{R}$ denotes the α -quantile of the recourse cost. Then, we have $\text{CVaR}_{\alpha}(Q(\mathbf{x}, \xi(\omega))) = \inf_{\eta \in \mathbb{R}} \{\eta + (1/(1-\alpha))v(\mathbf{x}, \eta)\}$. Since $E(\max\{Z, 0\})$ is a convex non-decreasing function in $Z \in \mathcal{Z}$ and $Q(\mathbf{x}, \xi(\omega)) - \eta$ is convex in $\mathbf{x}$ and η , the composite function $v(\mathbf{x}, \eta)$ is also convex in $\mathbf{x}$ and η . Therefore, it can be approximated by the subgradient inequalities. Using the chain rule of subdifferentiation we can calculate a subgradient of $v(\mathbf{x}, \eta)$ for given subgradients of $Q(\mathbf{x}, \xi(\omega)) - \eta$. Next, we present the master

problem based on the subgradients of $v(\mathbf{x}, \eta)$:

$$\min (1 + \lambda) \mathbf{c}^T \mathbf{x} + \theta_1 + \lambda \left(\eta + \frac{1}{1 - \alpha} \theta_2 \right) \quad (18)$$

$$\text{subject to } \mathbf{x} \in X, \quad (19)$$

$$\text{optimality cuts } \theta_1 \geq \sum_{s=1}^N p^s (\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}), \quad t = 1, \dots, \tau, \quad (20)$$

$$\text{optimality cuts } \theta_2 \geq v(\mathbf{x}^t, \eta^t) + (v_{\mathbf{x}}^t)^T (\mathbf{x} - \mathbf{x}^t) + (v_{\eta}^t)^T (\eta - \eta^t), \quad t = 1, \dots, \tau, \quad (21)$$

$$\eta \in \mathbb{R}, \quad (22)$$

$$\text{feasibility cuts } (T^s)^T \sigma^{gs} \mathbf{x} - (\mathbf{h}^s)^T \sigma^{gs} \geq 0, \quad s = 1, \dots, N, g_s = 1, \dots, G_s. \quad (23)$$

Observe that (21) and (22) replace the constraints (10)–(13) and the objective function representation is modified so that θ_2 represents $E([Q(\mathbf{x}, \xi(\omega)) - \eta]_+)$ instead of $\text{CVaR}_\alpha(Q(\mathbf{x}, \xi(\omega)))$. In this formulation, $\mathbf{x}^t$ and η^t denote the values of the variables $\mathbf{x}$ and η in the optimal solution of the master problem, respectively, at the iteration in which the t th optimality cut is constructed. $v_{\mathbf{x}}^t$ and v_{η}^t denote a subgradient of $v(\mathbf{x}, \eta)$ with respect to $\mathbf{x}$ and η , respectively. In particular, the subgradients used in the algorithm are given by

$$v_{\mathbf{x}}^t = \sum_{s=1}^N p^s \pi^{st} \mathbf{q}^{st} \quad \text{and} \quad v_{\eta}^t = - \sum_{s=1}^N p^s \varrho^{st}, \quad (24)$$

where, for all $s = 1, \dots, N$, $\pi^{st} = -(T^s)^T \mathbf{u}^{st}$, $\varrho^{st} = 1$ if $Q(\mathbf{x}^t, \xi^s) - \eta^t > 0$, and 0 otherwise.

As discussed for Algorithm 1, we can approximate $\mathbb{E}(Q(\mathbf{x}, \xi(\omega))) + \lambda \text{CVaR}_\alpha(Q(\mathbf{x}, \xi(\omega)))$ alternatively by introducing $\hat{\theta} = \theta_1 + \lambda(\eta + (1/(1-\alpha))\theta_2)$ and the combined optimality cuts:

$$\hat{\theta} \geq \sum_{s=1}^N p^s (\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}) + \lambda \left(\eta^t + \frac{1}{1 - \alpha} (v(\mathbf{x}^t, \eta^t) + (v_{\mathbf{x}}^t)^T (\mathbf{x} - \mathbf{x}^t) + (v_{\eta}^t)^T (\eta - \eta^t)) \right), \quad t = 1, \dots, \tau. \quad (25)$$

Algorithm 2. Decomposition Algorithm 2.

- 1: Step 1 of Algorithm 1.
- 2: Solve the master problem (18)–(23).
Let $(\bar{\mathbf{x}}, \bar{\eta}, \bar{\theta}_1, \bar{\theta}_2)$ be the optimal solution and $\bar{\theta} = \bar{\theta}_1 + \lambda(\bar{\eta} + \frac{1}{1-\alpha}\bar{\theta}_2)$.
- 3: Step 3 of Algorithm 1.
- 4: Step 4 of Algorithm 1.
- 5: Let $(\mathbf{x}^\tau, \eta^\tau) = (\bar{\mathbf{x}}, \bar{\eta})$.
- 6: Calculate the mean-risk function value of the recourse cost at the current solution:
$$\theta^* = \sum_{s=1}^N p^s (\mathbf{u}^{s\tau})^T (\mathbf{h}^s - T^s \mathbf{x}^\tau) + \lambda \left(\eta^\tau + \frac{1}{1-\alpha} v(\mathbf{x}^\tau, \eta^\tau) \right),$$

where $v(\mathbf{x}^\tau, \eta^\tau) = \sum_{s=1}^N p^s [(\mathbf{u}^{s\tau})^T (\mathbf{h}^s - T^s \mathbf{x}^\tau) - \eta^\tau]_+$.
- 7: **if** $\bar{\theta} \geq \theta^*$ **then**
- 8: Stop. $\mathbf{x}^\tau$ is the optimal first-stage decision vector.
- 9: **else**
- 10: Introduce the optimality cuts (20)–(21) for $\mathbf{u}^{s\tau}$ and $(\mathbf{x}^\tau, \eta^\tau)$.
For the second type of optimality cuts (21), calculate the subgradients as in (24). Go to Step 2.
- 11: **end if**

The description of this subgradient-based decomposition algorithm is presented in Algorithm 2. As in Algorithm 1, we consider a stopping criterion based on a specified tolerance value ε . The first algorithm stops after generating a smaller number of optimality cuts with respect to Algorithm 2, as presented in Section 6.2. However, one might expect that the subgradient approach takes generally less time even if it requires a larger number of optimality cuts, since it involves a significantly less number of variables. We provide comparative results on the computational performance of the proposed algorithms in Section 6.2.

3.3. The multicut versions of decomposition algorithms

In the decomposition algorithms presented in Sections 3.1 and 3.2, when the second-stage problem is feasible for all the realizations of the random data at an iteration, all the second-stage problems are solved to optimality and the information represented by the optimal dual variables is aggregated to generate the optimality cuts. At each such iteration of Algorithms 1 and 2 we generate a single-cut (9) or (20), respectively, for approximating the expectation term and a single-cut (10) or (21), respectively, for the CVaR term. Alternatively, in Algorithms 1 and 2, a single combined optimality cut (17) or (25), respectively, is generated to approximate the mean-risk function of the recourse cost. In a multicut method, instead of these aggregated cuts, one cut per realization is generated to keep more information about the dual solutions of the second-stage problems.

In Sections 3.1 and 3.2, we use the variables θ_1, θ_2 and $\hat{\theta}$ to define the optimality cuts. In the multicut method, we introduce the variables θ_1^s, θ_2^s , and $\hat{\theta}^s$ to generate the optimality cuts associated with the realization vector $\xi^s, s = 1, \dots, N$. Next, we specify the multicut versions of the optimality cuts introduced in Sections 3.1 and 3.2.

• For Algorithm 1

- The multicut version of the optimality cuts (9):

$$\theta_1^s \geq (\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}), \quad t = 1, \dots, \tau, s = 1, \dots, N. \quad (26)$$

- The multicut version of the optimality cuts (10):

$$\theta_2^s \geq \eta^t + \frac{1}{1 - \alpha} v^{st}, \quad t = 1, \dots, \tau, s = 1, \dots, N. \quad (27)$$

- The multicut version of the combined optimality cut (17):

$$\hat{\theta}^s \geq (\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}) + \lambda \left(\eta^t + \frac{1}{1 - \alpha} v^{st} \right), \quad t = 1, \dots, \tau, s = 1, \dots, N. \quad (28)$$

• For Algorithm 2

- The multicut version of the optimality cuts (20):

$$\theta_1^s \geq (\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}), \quad t = 1, \dots, \tau, s = 1, \dots, N. \quad (29)$$

- The multicut version of the optimality cuts (21):

$$\theta_2^s \geq v^s(\mathbf{x}^t, \eta^t) + (v_{\mathbf{x}}^{st})^T (\mathbf{x} - \mathbf{x}^t) + (v_{\eta}^{st})^T (\eta - \eta^t), \quad t = 1, \dots, \tau, s = 1, \dots, N, \quad (30)$$

where $v^s(\mathbf{x}^t, \eta^t) = [Q(\mathbf{x}^t, \xi^s) - \eta^t]_+$, $v_{\mathbf{x}}^{st} = -(T^s)^T \mathbf{u}^{st} \varrho^{st}$, $v_{\eta}^{st} = -\varrho^{st}$, and $\varrho^{st} = 1$ if $Q(\mathbf{x}^t, \xi^s) - \eta^t > 0$, and 0 otherwise.

- The multicut version of the combined optimality cut (25):

$$\hat{\theta}^s \geq (\mathbf{u}^{st})^T (\mathbf{h}^s - T^s \mathbf{x}) + \lambda \left(\eta^t + \frac{1}{1 - \alpha} (v^s(\mathbf{x}^t, \eta^t) + (v_{\mathbf{x}}^{st})^T (\mathbf{x} - \mathbf{x}^t) + (v_{\eta}^{st})^T (\eta - \eta^t)) \right), \quad (31)$$

$$t = 1, \dots, \tau, s = 1, \dots, N.$$

In addition to the multicut versions of the optimality cuts, we replace θ_1 by $\sum_{s=1}^N p^s \theta_1^s$, θ_2 by $\sum_{s=1}^N p^s \theta_2^s$, and $\hat{\theta}$ by $\sum_{s=1}^N p^s \hat{\theta}^s$ in the objective functions of the master problems. Using the disaggregate cuts, as listed above, is generally expected to improve the outer approximations and reduce the number of iterations with respect to the single-cut method. However, the size of the master problem may cause computational difficulties, since we add a large number of optimality cuts at once. Note that in the single-cut versions of the decomposition algorithms, the number of optimality cuts is equal to 2τ . In the multicut version of Algorithm 1 with the combined optimality cuts (28), the number of optimality cuts is equal to $N\tau$. When one type of cuts ((26) or (27)) and both types of cuts (26) and (27) are introduced, the number of optimality cuts generated in Algorithm 1 is equal to $N\tau + \tau$ and $2N\tau$, respectively. Similarly, the number of optimality cuts generated in Algorithm 2 with (31), (29) or (30), (29) and (30) are $N\tau$, $N\tau + \tau$, and $2N\tau$, respectively. Note that in the risk-neutral two-stage stochastic linear programming we only have optimality cuts of type (26). To the best of our knowledge, Miller and Ruszczyński [7] is the only study implementing the multicut approach for the risk-averse two-stage stochastic programming. We extend the existing multicut methods by considering alternate options to introduce the multicut versions of the optimality cuts.

Birge and Louveaux [2] note that the multicut method is expected to be more effective when the number of realizations is not significantly larger than the number of first-stage constraints. The authors also remark that the trade-off between the number of iterations and the size of the master problem is instance-dependent. For a detailed discussion on the relative advantages and disadvantages of the single-cut and multicut methods we refer to Birge and Louveaux [28] and Gassmann [29]. As suggested by Birge and Louveaux [2] one may employ an approach that is somehow between the single-cut and the multicut methods. Such an approach groups all the realizations into disjoint subsets and aggregates the dual variable information over each group. In the single-cut method, there is only one group consisting of all the realizations and in the multicut method, each realization constitutes one group. In our numerical study, we provide a sample set of results for the multicut versions of Algorithm 2 in order to demonstrate how the multicut approach affects the computational performance.

In this study, we have focused on the cutting plane-based decomposition approaches to deal with a large number of scenarios when we consider CVaR as a risk measure in the two-stage mean-risk analysis. The proposed methods may further be improved computationally using regularization and bunching methods. The interested reader is referred to Ruszczyński [30], Ruszczyński and Świetanowski [31] and Birge and Louveaux [2]. We also refer to Ruszczyński [32] and Kall and Mayer [33] for alternative decomposition methods in two-stage stochastic programming.

4. The value of information and the stochastic solution for the two-stage mean-risk stochastic programming models

Stochastic programming approach to model decision making problems in the presence of uncertainty is known to be computationally more challenging than simply considering the underlying deterministic versions of the problems. Therefore, some decision makers may prefer to solve simpler deterministic versions instead of a stochastic programming problem. For example, such a decision maker can solve the deterministic problem where all the random parameters are assumed to take their expected values or several deterministic programs, each of which,

corresponds to one particular scenario, and combine the obtained different solutions by some heuristic rule. A natural question is whether solving such simpler versions of the problem provides reasonably good solutions and it is crucial to quantify the solution quality. The EVPI and the VSS are two well-known measures, which are designed to indicate whether it is worth to model the randomness using a risk-neutral stochastic programming approach. In other words, these measures are used to check whether the way the stochastic programming model incorporates the randomness has a significant effect on the obtained optimal solutions. However, note that these measures are based on the assumption that the set of scenarios used to model the randomness represents the stochastic nature of the problem parameters accurately. In this section, we first discuss the concepts of EVPI and VSS that are defined for the risk-neutral stochastic programming models. Then, we adapt these concepts for the proposed two-stage mean-risk stochastic programming model and define new measures on the VPI and VSS.

In the stochastic programming setting where the randomness is characterized by a set of scenarios, the EVPI is calculated by solving a family of problems, each associated with one particular scenario. These individual problems are known as the wait-and-see (WS) problems. Basically, this approach is based on the perfect information about the future. The WS models are often used to quantify how much a decision maker would be willing to pay to get complete information about the future realizations of the random data. Let $\bar{\mathbf{x}}(\xi^s)$ denote the optimal solution of the WS problem associated with the realization vector ξ^s , $s = 1, \dots, N$. Then, we have

$$\bar{\mathbf{x}}(\xi^s) = \arg \min_{\mathbf{x} \in X} f(\mathbf{x}, \omega^s) = \arg \min_{\mathbf{x} \in X} \{\mathbf{c}^T \mathbf{x} + Q(\mathbf{x}, \xi^s)\}, \quad s = 1, \dots, N.$$

Assume that we can find $\bar{\mathbf{x}}(\xi^s)$, and so the associated objective functions $f(\bar{\mathbf{x}}(\xi^s), \omega^s)$, $s = 1, \dots, N$. According to this WS approach, $f(\bar{\mathbf{x}}(\xi^s), \omega^s)$, $s = 1, \dots, N$, represent the realized optimal values of the objective function $f(\mathbf{x}, \omega)$. Using these realizations we can compute the expected optimal objective function value, which is known as the WS solution:

$$WS = E\left(\min_{\mathbf{x} \in X} f(\mathbf{x}, \omega)\right) = \sum_{s=1}^N p^s f(\bar{\mathbf{x}}(\xi^s), \omega^s). \quad (32)$$

Recall that the risk-neutral two-stage stochastic programming problem is in the following form:

$$\min_{\mathbf{x} \in X} \{\mathbb{E}(f(\mathbf{x}, \omega))\}.$$

This problem is also known as the recourse problem or the here-and-now problem and the optimal objective function value is denoted by RP. Then by definition, the EVPI is calculated as follows: EVPI = RP – WS. Note that the WS approach provides a set of solutions instead of one solution that would be implementable. Due to this difficulty, alternatively it has been proposed to solve a much simpler problem known as the expected value problem:

$$EV = \min_{\mathbf{x} \in X} \{\mathbf{c}^T \mathbf{x} + Q(\mathbf{x}, \bar{\xi})\},$$

where $\bar{\xi} = E(\xi(\omega))$. Let us denote the optimal solution of the expected value problem by $\bar{\mathbf{x}}(\bar{\xi})$. For the expected value solution $\bar{\mathbf{x}}(\bar{\xi})$ we can solve the second-stage problem associated with each scenario and obtain the following set of realized values of the objective function $f(\mathbf{x}, \omega)$:

$$f(\bar{\mathbf{x}}(\bar{\xi}), \omega^s) = \mathbf{c}^T \bar{\mathbf{x}}(\bar{\xi}) + Q(\bar{\mathbf{x}}(\bar{\xi}), \xi^s), \quad s = 1, \dots, N.$$

In the EV model, we basically handle the uncertainty without incorporating it into the underlying optimization model. This is

basically a naive risk-neutral approach and how bad the decision $\bar{\mathbf{x}}(\xi)$ performs is measured by the VSS: $VSS = EEV - RP$. Here EEV denotes the expected result of using the EV solution:

$$EEV = E(f(\bar{\mathbf{x}}(\xi), \omega)) = \sum_{s=1}^N p^s f(\bar{\mathbf{x}}(\xi), \omega^s). \quad (33)$$

After discussing the concepts of the EVPI and VSS, we would like to emphasize that these measures cannot be directly used to argue whether it is worth to solve the proposed risk-averse two-stage stochastic programming problem. The measures defined above are based on the expected values and so they are useful to evaluate the risk-neutral stochastic programming models. Therefore, we define new measures on the VPI and the VSS in order to quantify the effect of involving also the risk term in the two-stage stochastic model. Basically, instead of computing the expected value of the optimal objective function value in (32) and the expected result of using the EV solution in (33), we propose to calculate the mean-risk function value associated with the realizations of the objective function $f(\mathbf{x}, \omega)$ obtained by the WS and the EV approaches, respectively. These new stochastic measures provide more information about the probability distributions of the objective function values associated with the WS and EV decisions. More precisely, for the risk parameters λ and α we define the stochastic measures for the proposed two-stage mean-risk model involving CVaR as follows:

$$MRRP = \min_{\mathbf{x} \in X} \{E(f(\mathbf{x}, \omega)) + \lambda \text{CVaR}_\alpha(f(\mathbf{x}, \omega))\},$$

$$MRWS = E\left(\min_{\mathbf{x} \in X} f(\mathbf{x}, \omega)\right) + \lambda \text{CVaR}_\alpha\left(\min_{\mathbf{x} \in X} f(\mathbf{x}, \omega)\right),$$

$$MRVPI = MRRP - MRWS,$$

$$MREV = E(f(\bar{\mathbf{x}}(\xi), \omega)) + \lambda \text{CVaR}_\alpha(f(\bar{\mathbf{x}}(\xi), \omega))$$

$$MRVSS = MREV - MRRP.$$

The EVPI represents the gain of perfect information on average, whereas $MRVPI$ measures the gain of perfect information based on the mean-risk function value corresponding to the objective function values obtained from the WS decisions. Thus, it quantifies the value of information based on the specified risk preferences. The VSS measures the cost of ignoring uncertainty on average while making a decision. However, it only focuses on the risk-neutral approach. The new measure $MRVSS$ is defined to measure the possible gain from solving the stochastic model that incorporates the mean-risk function for the random outcome of interest into the problem formulation. Note that the high values of $MRVSS$ would indicate the significant improvement in the solution quality in terms of the specified risk preference by solving the risk-averse model instead of solving the expected value problem.

Similar to the risk-neutral case, we can obtain basic relations between $MRRP$, $MRWS$ and $MREV$, as stated in the following proposition. Note that a risk measure ρ is said to have the monotonicity property if $\rho(X_1) \geq \rho(X_2)$ for any $X_1, X_2 \in \mathcal{Z}$ such that $X_1(\omega) \geq X_2(\omega)$ for all $\omega \in \Omega$ (see, e.g., [34]). Let us denote the optimal solution associated with the $MRRP$ by $\mathbf{x}^*$.

Proposition 4.1. *For any two-stage mean-risk stochastic programming problem (3), the following relations hold true:*

$$MRWS \leq MRRP \leq MREV.$$

Thus, $MRVPI \geq 0$ and $MRVSS \geq 0$.

Proof. Since in the WS approach, an optimal solution $\bar{\mathbf{x}}(\xi(\omega))$ is obtained for each realization of $\xi(\omega)$, we have $f(\bar{\mathbf{x}}(\xi(\omega)), \omega) \leq f(\mathbf{x}^*, \omega)$ for all $\omega \in \Omega$. Calculating the mean-risk function value of

both sides for any $\lambda \geq 0$ yields the first inequality due to the monotonicity of the expectation operator and CVaR. The EV problem provides us with the optimal solution $\bar{\mathbf{x}}(\xi)$, which is just a feasible solution of the risk-averse recourse problem. Therefore, the mean-risk function value associated with $\bar{\mathbf{x}}(\xi)$, denoted by $MREV$, is larger than or equal to $MRRP$. $\square$

Since in this study we specify the CVaR as the risk measure, we only present the stochastic measures involving CVaR. However, it is easy to see that these definitions and the relations in Proposition 4.1 are valid for any risk function with the convexity and monotonicity properties. For some examples of such risk functions, we refer the reader to Ogryczak and Ruszczyński [20] and Ruszczyński and Shapiro [35].

5. Application to disaster preparedness management

We apply the proposed framework to a disaster preparedness management problem, where modeling the variability of random outcomes is crucial in order to reduce the damage of a possible disaster. We consider an existing disaster location-allocation model proposed by Rawls and Turnquist [16]. The aim of their model is to determine the response facility locations and the inventory levels of several commodities at each facility in the presence of uncertainty in demand, transportation capacities, and the damage level of supplies. This is a significant model, which considers different types of uncertainties. In particular, the model allows the links in the network and some or all of the supplies preallocated at a facility to be damaged when the disaster occurs. We intend to contribute to the disaster management literature by extending this existing model by incorporating a risk measure. Our proposed model is a significant and non-trivial extension, because it incorporates a risk measure in addition to the expectation criterion. After briefly presenting the existing model we describe our proposed model and finally, we discuss some details regarding the application of the decomposition algorithms.

5.1. Existing model

The problem of determining the locations of the response facilities and the inventory levels of disaster relief supplies at each facility is crucial to effectively manage the response operations when a disaster occurs. A limited number of studies exist in the literature, which focus on the pre-positioning of the different types of emergency supplies in the presence of uncertainty. We refer to Rawls and Turnquist [16] and the references therein for positioning of this model in the disaster management literature.

In two-stage programming problems there is a hierarchy of decisions. In the first-stage, a set of decisions is made and then uncertainty is revealed, i.e., random variables are observed, and then a second set of decisions are determined under the pre-determined first-stage decision variables and the observed uncertain quantities. This setup fits the problem of pre-positioning emergency supplies at a stochastic disaster relief network perfectly. The disaster relief supplies are preallocated at the response facilities which are distributed according to the realized demand in response to a disaster. In the proposed two-stage model, the first-stage decisions are on the types and locations of the facilities, and on the amount of each commodity to be stocked at each facility. For example, the commodities that are likely to be in demand when a hurricane occurs are prepackaged food, medical kits, blankets, etc. The second-stage decision variables are concerned with distributing the supplies to satisfy the realized demand. The second-stage problem, which is in the form of a multi-commodity network flow formulation, involves stochastic

constraints to satisfy the demand. Satisfying the stochastic constraints for all the realizations of random demand values would lead to overly conservative decisions. Instead, we allow infeasibilities in the stochastic constraints, but introduce penalties on the amount of their violations. Therefore, auxiliary second-stage variables are introduced to calculate the surplus and shortage amounts to control the salvage costs and penalties on unmet demand.

We basically use almost the same notation introduced in Rawls and Turnquist [16].

Inputs

- K , set of commodities, indexed by k ;
- I , set of locations (nodes), indexed by i ;
- A , set of arcs (links) in the network;
- L , size categories of a storage facility, indexed by l ;
- M_l , the overall capacity (e.g., square feet of available space) of a facility of size category l , $l \in L$;
- F_{il} , fixed cost of opening a facility of size category l at location i , $i \in I$, $l \in L$;
- b^k , unit space requirement for commodity k , $k \in K$;
- q^k , unit purchasing cost for commodity k , $k \in K$;
- hc^k , unit salvage cost for commodity k (when there is surplus), $k \in K$;
- pc^k , unit penalty cost for the shortage of commodity k , $k \in K$;
- u^k , link capacity used for shipping a unit of commodity k , $k \in K$;
- N , number of scenarios, indexed by s ;
- p^s , probability of scenario s , $s = 1, \dots, N$;
- c_{ij}^{ks} , cost of shipping a unit of commodity k through the link (i, j) under scenario s , $(i, j) \in A$, $k \in K$, $s = 1, \dots, N$;
- g_i^{ks} , realization of demand for commodity k at location i under scenario s , $i \in I$, $k \in K$, $s = 1, \dots, N$;
- γ_i^{ks} , realization of the proportion of commodity k at location i that remains usable under scenario s ($0 \leq \gamma_i^{ks} \leq 1$), $i \in I$, $k \in K$, $s = 1, \dots, N$;
- U_{ij}^s , realization of the available capacity of link (i, j) under scenario s (it may be even zero), $(i, j) \in A$, $s = 1, \dots, N$.

Decision variables

First-stage variables:

- r_{ik} , amount of commodity k pre-positioned at location i ,
- $y_{il} = \begin{cases} 1 & \text{if a facility of size category } l \text{ is located at node } i, \\ 0 & \text{otherwise.} \end{cases}$

Primary second-stage variables:

- x_{ij}^{ks} , amount of commodity k shipped through link (i, j) under scenario s ,

Auxiliary second-stage variables:

- z_i^{ks} , amount of commodity k that is not used at location i under scenario s (surplus amount),
- w_i^{ks} , amount of the shortage of commodity k at location i under scenario s .

The randomness in the disaster network is modeled using a set of scenarios. Note that a scenario represents the joint realization of demand for each commodity (g_i^{ks} , $i \in I$, $k \in K$), the arc capacities (U_{ij}^s , $(i, j) \in A$), and the damage levels of supplies (γ_i^{ks} , $i \in I$, $k \in K$). Without loss of generality, we assume that I is also the set of candidate storage facilities. For a first-stage decision vector $(\mathbf{r}, \mathbf{y})$

the objective function of the first-stage problem is given next:

$$f(\mathbf{r}, \mathbf{y}, \omega) = \sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{k \in K} \sum_{i \in I} q^k r_i^k + Q(\mathbf{r}, \mathbf{y}, \xi(\omega)),$$

where $\omega \in \Omega$ and the vector $(\mathbf{r}, \mathbf{y})$ satisfies the following constraints:

$$\sum_{k \in K} b^k r_i^k \leq \sum_{l \in L} M_l y_{il}, \quad i \in I, \quad (34)$$

$$\sum_{l \in L} y_{il} \leq 1, \quad i \in I, \quad (35)$$

$$y_{il} \in \{0, 1\}, \quad i \in I, l \in L, \quad (36)$$

$$r_i^k \geq 0, \quad i \in I, k \in K. \quad (37)$$

For the following second-stage problem, $\xi^s = (\mathbf{c}^s, \gamma^s, \mathcal{G}^s, \mathbf{U}^s)$ denotes the realization of the random data under scenario s , $s \in \{1, \dots, N\}$:

$$Q(\mathbf{r}, \mathbf{y}, \xi^s) = \min \sum_{(i, j) \in A} \sum_{k \in K} c_{ij}^{ks} x_{ij}^{ks} + \sum_{i \in I} \sum_{k \in K} (hc^k z_i^{ks} + pc^k w_i^{ks}) \quad (38)$$

$$\begin{aligned} \text{subject to } & \sum_{(j, i) \in A} x_{ji}^{ks} + \gamma_i^{ks} r_i^k - \sum_{(i, j) \in A} x_{ij}^{ks} - g_i^{ks} = z_i^{ks} - w_i^{ks}, \\ & i \in I, k \in K, \end{aligned} \quad (39)$$

$$\sum_{k \in K} u^k x_{ij}^{ks} \leq U_{ij}^s, \quad (i, j) \in A, \quad (40)$$

$$x_{ij}^{ks} \geq 0, \quad (i, j) \in A, k \in K, \quad (41)$$

$$z_i^{ks}, w_i^{ks} \geq 0, \quad i \in I, k \in K. \quad (42)$$

Constraints (34) guarantee that commodities are stored at the open facilities and the total space taken by the pre-allocated supplies does not exceed the capacity of the facility. The number of facilities to be located at any node is at most 1 by constraints (35). The conservation of flow at each node is represented by constraints (39). By dropping the scenario index, we can write the stochastic version of the demand satisfaction constraint as follows:

$$\sum_{(j, i) \in A} x_{ji}^k(\omega) - \sum_{(i, j) \in A} x_{ij}^k(\omega) + \gamma_i^k(\omega) r_i^k - g_i^k(\omega) = z_i^k(\omega) - w_i^k(\omega), \quad i \in I.$$

Notice that the random amount of available commodity k at node i is equal to $a^k(i) = \sum_{(j, i) \in A} x_{ji}^k(\omega) - \sum_{(i, j) \in A} x_{ij}^k(\omega) + \gamma_i^k(\omega) r_i^k$, where $\gamma_i^k(\omega) r_i^k$ is the amount of undamaged supplies that are prelocated at node i . If the realized random amount of commodity k required at node i under scenario s , g_i^{ks} , is larger than the realization of $a^k(i)$ under scenario s , we have a positive shortage $w_i^{ks} > 0$, otherwise we have a positive surplus $z_i^{ks} > 0$. Constraints (40) enforce that the assigned link flow does not exceed the available capacity. The rest of the constraints are for the non-negativity and binary restrictions.

In the existing model proposed by Rawls and Turnquist [16], the expectation is used as the preference criterion: the existing model finds the location-allocation policy vector $(\mathbf{y}, \mathbf{r})$ for which the associated random total cost $f(\mathbf{r}, \mathbf{y}, \omega)$ has the minimum expected value. Rawls and Turnquist [16] conclude their paper by emphasizing the significance of involving risk measures into their model and indicate that studying appropriate modifications in the objective function is a desirable future research direction. In the next section, we extend their model by considering CVaR of the total cost and present the proposed risk-averse model.

5.2. Proposed model

As discussed in Section 2, we specify CVaR as the risk measure and formulate the two-stage disaster preparedness model as a two-stage mean-risk stochastic programming problem:

$$\min_{(\mathbf{r}, \mathbf{y}) \in X} \{ \mathbb{E}(f(\mathbf{r}, \mathbf{y}, \omega)) + \lambda \text{CVaR}_\alpha(f(\mathbf{r}, \mathbf{y}, \omega)) \},$$

where the feasible set X and the second-stage problem are defined by (34–37) and (38–42), respectively.

By Proposition 2.1 we have the deterministic equivalent formulation (DEF) of the stochastic disaster location–allocation problem in the following form:

$$\begin{aligned} \min \quad & (1 + \lambda) \left(\sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{k \in K} \sum_{i \in I} q c^k r_i^k \right) \\ & + \sum_{s=1}^N p^s \left(\sum_{(i,j) \in A} c_{ij}^{ks} x_{ij}^{ks} + \sum_{i \in I} \sum_{k \in K} (h c^k z_i^{ks} + p c^k w_i^{ks}) \right) \\ & + \lambda \left(\eta + \frac{1}{1-\alpha} \sum_{s=1}^N p^s v^s \right) \\ \text{subject to} \quad & \sum_{(i,j) \in A} x_{ji}^{ks} - \sum_{(i,j) \in A} x_{ij}^{ks} - z_i^{ks} + w_i^{ks} = g_i^{ks} - \gamma_i^{ks} r_i^k, \\ & i \in I, k \in K, s = 1, \dots, N, \\ & \sum_{k \in K} u^k x_{ij}^{ks} \leq U_{ij}^s, \quad (i,j) \in A, s = 1, \dots, N, \\ & \sum_{k \in K} b^k r_i^k \leq \sum_{l \in L} M_l y_{il}, \quad i \in I, \\ & \sum_{l \in L} y_{il} \leq 1, \quad i \in I, \\ & y_{il} \in \{0, 1\}, \quad i \in I, l \in L, \\ & r_i^k \geq 0, \quad i \in I, k \in K, \\ & x_{ij}^{ks} \geq 0, \quad (i,j) \in A, k \in K, s = 1, \dots, N, \\ & z_i^{ks}, w_i^{ks} \geq 0, \quad i \in I, k \in K, s = 1, \dots, N, \\ & v^s \geq \sum_{(i,j) \in A} \sum_{k \in K} c_{ij}^{ks} x_{ij}^{ks} + \sum_{i \in I} \sum_{k \in K} (h c^k z_i^{ks} + p c^k w_i^{ks}) - \eta, \quad s = 1, \dots, N, \\ & \eta \in \mathbb{R}, \quad v^s \geq 0, s = 1, \dots, N. \end{aligned}$$

We refer to this proposed problem with CVaR as “disaster location and allocation problem with CVaR (DALPWithCVaR)”. By changing the trade-off coefficient λ , the efficient location and allocation policies can be constructed, and this would allow the decision makers to evaluate different policies. Illustrative examples are presented in our numerical section.

In the first stage of the proposed disaster management model there exist binary variables, $y_{il}, i \in I, l \in L$, to prescribe the locations of the facilities. However, the proposed framework, which has been described for continuous variables only, can still be applied under these integrality restrictions. Observe that the binary variables do not appear in the second-stage problem (38)–(42) and therefore, the recourse function depends on $\mathbf{y}$ through the continuous variables $\mathbf{r}$. Due to this special structure, we can just drop the binary variables from the recourse function and then we can rewrite the mean-risk function of the model as

$$(1 + \lambda) \left(\sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{k \in K} \sum_{i \in I} q c^k r_i^k \right) + \mathbb{E}(Q(\mathbf{r}, \xi(\omega))) + \lambda \text{CVaR}_\alpha(Q(\mathbf{r}, \xi(\omega))).$$

As discussed in Section 2, the mean-risk function of the recourse cost is convex in $\mathbf{r}$ and we can construct the outer approximations for it in the continuous variables $\mathbf{r}$. Thus, the decomposition algorithms described in Section 3 directly apply here. The only difference is that Step 2 of Algorithms 1 and 2 requires solving a mixed-integer programming master problem. One may

incorporate a heuristic into the decomposition algorithms to obtain a near-optimal solution to the master problem more quickly. To this end, a heuristic based on a Lagrangian relaxation has been proposed by Rawls and Turnquist [16]. Alternatively, one may utilize a branch-and-cut scheme in the master problem.

Next, we discuss how Algorithms 1 and 2 can be employed to solve DALPWithCVaR. In the second-stage problem, auxiliary variables guarantee the feasibility. Therefore, we have a complete recourse and we do not need to consider the feasibility cuts in the master problems of the decomposition algorithms. In Algorithm 1, the optimality cuts for DALPWithCVaR are generated as follows:

$$\begin{aligned} \theta_1 &\geq \sum_{s=1}^N p^s \left(\sum_{k \in K, i \in I} \beta_{ik}^{st} (g_i^{ks} - \gamma_i^{ks} r_i^k) + \sum_{(i,j) \in A} \mu_{ij}^{st} U_{ij}^s \right), \quad t = 1, \dots, \tau, \\ \theta_2 &\geq \eta^t + \frac{1}{1-\alpha} \sum_{s=1}^N p^s v^{st}, \quad t = 1, \dots, \tau, \\ v^{st} &\geq \sum_{k \in K, i \in I} \beta_{ik}^{st} (g_i^{ks} - \gamma_i^{ks} r_i^k) + \sum_{(i,j) \in A} \mu_{ij}^{st} U_{ij}^s - \eta^t, \quad s = 1, \dots, N, \quad t = 1, \dots, \tau, \\ v^{st} &\geq 0, \quad s = 1, \dots, N, t = 1, \dots, \tau, \end{aligned}$$

where β_{ik}^{st} and μ_{ij}^{st} are the optimal dual variables associated with the constraints (39) and (40) of the second-stage problem for realization ξ^s , respectively, at the iteration in which the t th optimality cut of each type is constructed.

In order to calculate the subgradients in Algorithm 2, we note that the component of π^{st} , $s = 1, \dots, N$, corresponding to the first-stage variable r_i^k is equal to $-\gamma_i^{ks} \beta_{ik}^{st}$ and all the components corresponding to the $\mathbf{y}$ variables are equal to 0, because the $\mathbf{y}$ variables do not occur in the second-stage problem.

6. Numerical results

In this section, we present numerical results for the case study constructed in Rawls and Turnquist [16] to show how the optimal location and allocation policies change with respect to the risk parameters. We also provide computational results for a number of problem instances to illustrate the computational effectiveness of the proposed decomposition algorithms.

The optimization problems are modeled with the AMPL mathematical programming language [36] running on the 11.2 CPLEX solver [37]. Each problem instance is solved on a single core of a 64-bit HP workstation running on Linux with 2 quad-core 1.6 GHz CPU, and 16 GB of RAM. All reported CPU times are in “seconds”.

6.1. Case study

We solve the proposed problem DALPWithCVaR for the case study presented in Rawls and Turnquist [16] considering different values of the risk parameters λ and α . Rawls and Turnquist [16] use the expected value as the preference criterion; their model is a special case of the proposed model when $\lambda = 0$. In this section, we present comparative results to show the effect of incorporating a risk measure on the random total cost and analyze how the policies change with respect to the risk parameters.

Here we briefly discuss the key points of the case study and refer the reader to Rawls and Turnquist [16] for further details. The authors consider a network consisting of 30 nodes and 55 links to represent the southeastern US. They focus on hurricane threats and construct 51 scenarios based on the historical characteristics of hurricanes in the southeastern region of US. In the case study, three emergency commodities are considered: water,

food and medical kits. Facilities are of three possible sizes: small, medium, and large. Food is considered in units of 1000 meals-ready-to eat (MREs), and water is considered in units of 1000 gal. It is assumed that the unit penalty cost pc^k for each commodity k is κ times the purchase price qc^k , $k \in K$. They set $\kappa = 10$, we also consider $\kappa = 5$ in our study. In Table 1, we provide the values of the parameters that might be useful to interpret the numerical results. The unmet demand penalty (shortage) costs are crucial while determining the policies, therefore, note that the shortage penalty cost is the highest for the food and the lowest for the medical kits.

In our proposed framework, there are two risk parameters: α and λ . Here we discuss how these risk parameters affect the optimal policies. The specified α level represents the risk preference in percentage terms, i.e., CVaR_α quantifies the mean value of the worst $(1-\alpha)\%$ of the total costs. When α increases the corresponding value-at-risk increases, and CVaR_α accounts for the risk of larger realizations. Thus, larger α values would lead to more conservative policies, which give more weight to worse scenarios. Note also that increasing the value of λ would increase the relative importance of the risk term and so would also lead to more risk-averse policies. Thus, increasing the parameter λ and/or the parameter α implies a higher level of risk

Table 1
Input parameters in the case study.

For commodities				For facility types			
Commodity	qc^k (\$/unit)	b^k (ft ³ /unit)	Transportation cost (\$/unit mile)	Size category	Description	F_i (\$)	M_i (ft ³)
Water (1000 gal)	647.7	144.6	0.3	1	Small	19,600	36,400
Food (1000 MREs)	5420	83.33	0.04	2	Medium	188,400	408,200
Medical kits	140	1.16	0.00058	3	Large	300,000	780,000

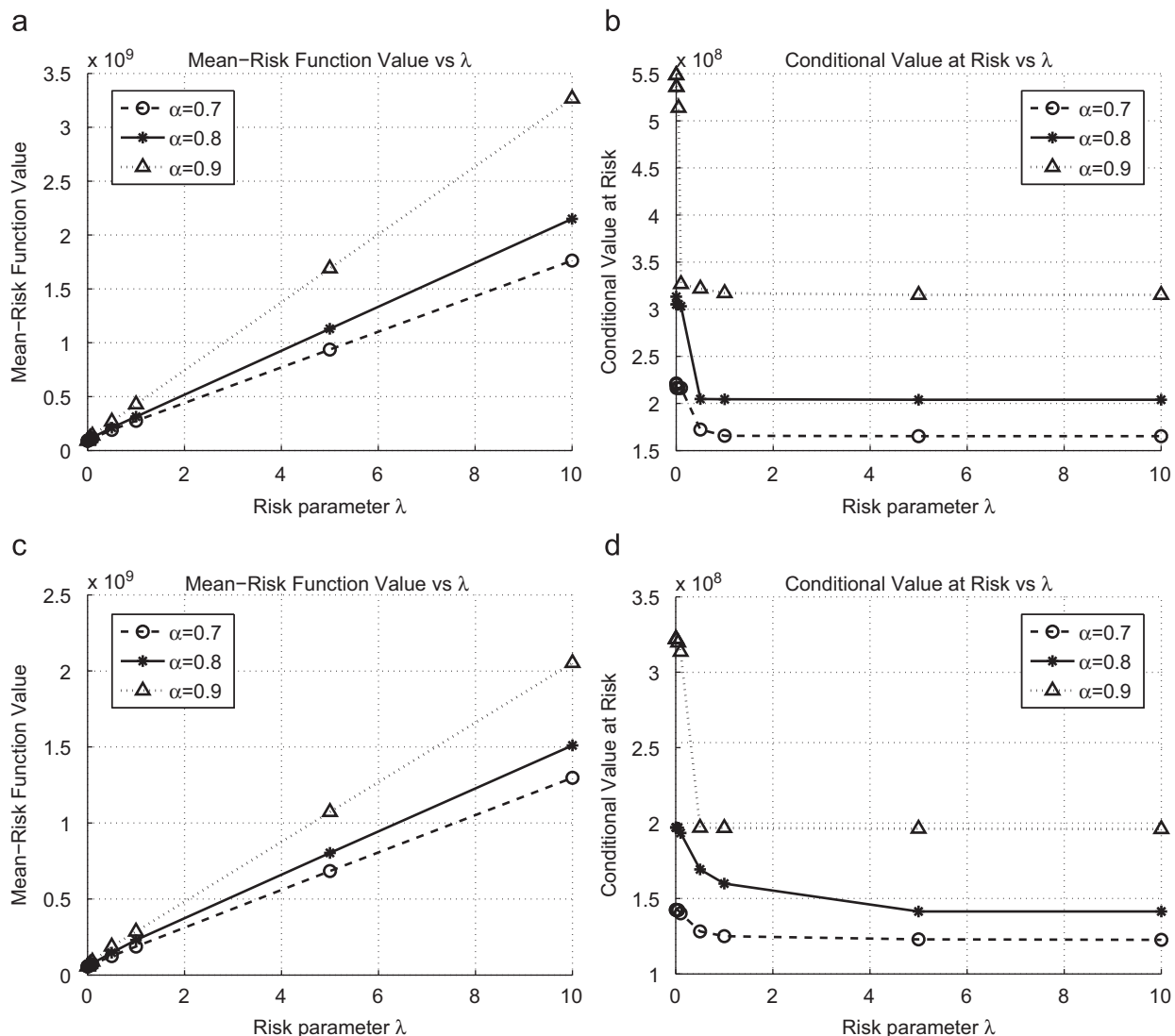


Fig. 1. Mean-risk function value of $f(\mathbf{r}, \mathbf{y}, \omega)$ and $\text{CVaR}_\alpha(f(\mathbf{r}, \mathbf{y}, \omega))$. (a,b) $\kappa = 10$. (c,d) $\kappa = 5$.

aversion. Fig. 1 illustrates how the (near-) optimal mean-risk function of the total cost increases as the risk parameters λ or α increase (see Fig. 1(a) and (c)). Similar to the (near-) optimal mean-risk

function, CVaR also increases as α increases by the definition of CVaR, i.e., for a larger value of α we focus on larger realizations of the total cost and the conditional expectation is larger as seen in

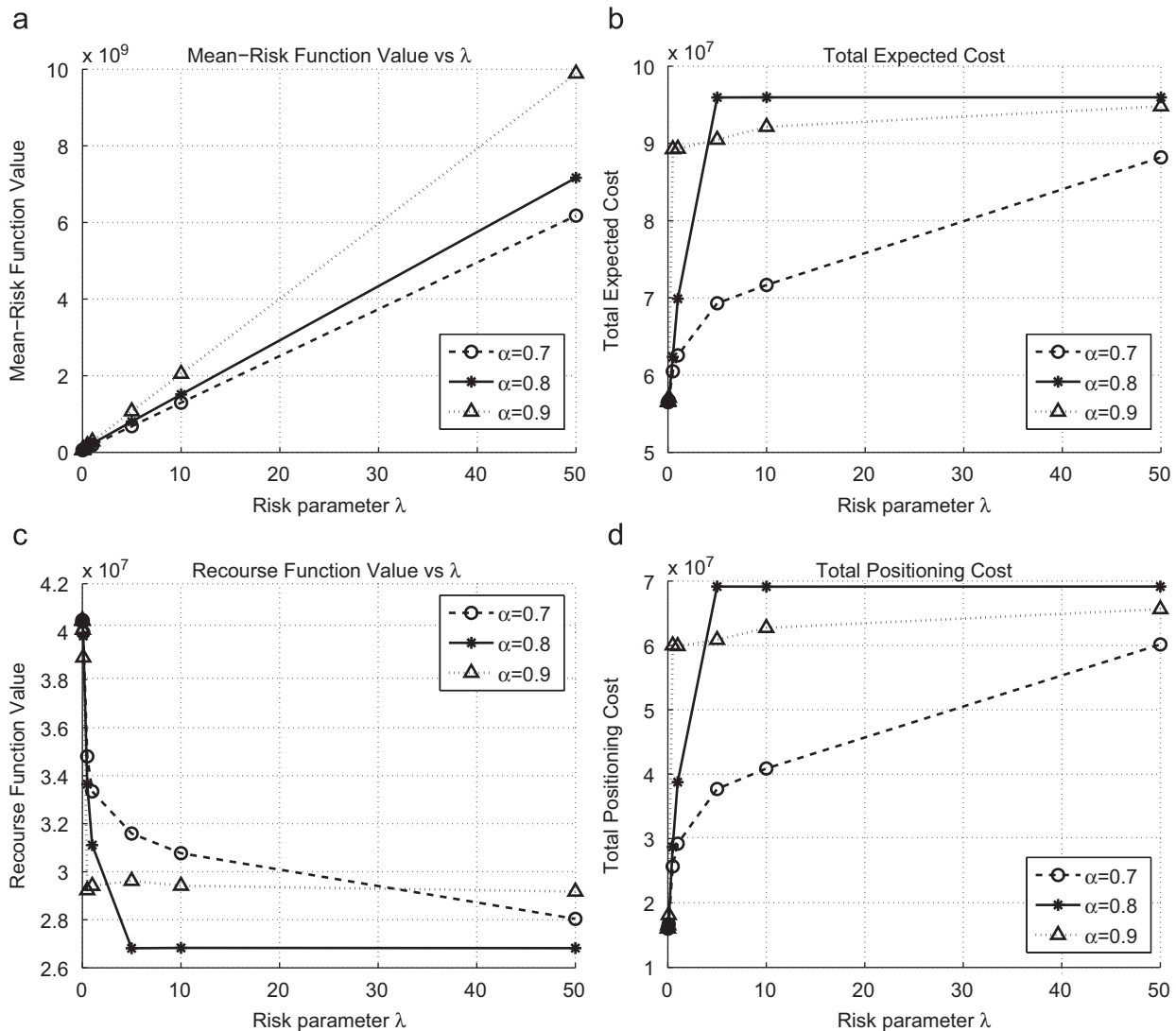


Fig. 2. Cost function values for different values of κ and λ parameters. (a–d) $\kappa = 5$.

Table 2

Detailed cost values, total inventory levels, total number of facilities, shortage probabilities for $\kappa = 5$.

α	Base problem, risk coefficient $\lambda = 0$		Risk coefficient $\lambda = 0.1$		Risk coefficient $\lambda = 1$	
	0.7	0.8	0.7	0.8	0.7	0.8
Mean-risk func.	56.51×10^6	56.51×10^6	70.71×10^6	76.13×10^6	187.57×10^6	229.74×10^6
CVaR $_{\alpha}$	142.44×10^6	197.09×10^6	140.18×10^6	193.43×10^6	125.01×10^6	159.85×10^6
Total exp. cost	56.51×10^6	56.51×10^6	56.70×10^6	56.78×10^6	62.56×10^6	69.88×10^6
Positioning cost	16.03×10^6	16.03×10^6	16.72×10^6	16.96×10^6	29.21×10^6	38.77×10^6
Exp. recourse func.	40.48×10^6	40.48×10^6	39.97×10^6	39.82×10^6	33.35×10^6	31.11×10^6
Exp. trans. and salvage cost	2.02×10^6	2.02×10^6	2.21×10^6	2.23×10^6	4.8×10^6	6.79×10^6
Exp. shortage cost	38.46×10^6	38.46×10^6	37.76×10^6	37.59×10^6	28.55×10^6	24.32×10^6
Water (1000 gal)	4920.1	4920.1	5659.9	5043.9	7924.9	8999
Food (1000 meals)	1830	1830	1933	2052	3921	5493.6
Medical kits	17,560.1	17,560.1	15,049	15,049	15,049	16,688.8
Small facilities	14	14	5	14	8	4
Medium facilities	1	1	2	1	3	4
Large facilities	0	0	0	0	0	0
Shortage prob. water	0.286	0.286	0.226	0.258	0.213	0.16
Shortage prob. food	0.235	0.235	0.198	0.186	0.098	0.079
Shortage prob. med. kits	0.363	0.363	0.41	0.41	0.465	0.416

Fig. 1(b) and (d). However, CVaR decreases as λ increases. Due to the changing trade-off between the expectation and the CVaR criterion, larger λ values provide us with a higher expected total cost and a lower CVaR value. However, we cannot make such a claim for the parameter α , i.e., a larger α value does not always result in a higher

expected total cost. Thus, for a fixed α it is easy to see how the cost values change with respect to the risk coefficient λ . However, for a fixed λ value, we can only claim that a larger α value leads to higher mean-risk function and CVaR values, but the expected cost value may decrease or increase. Fig. 2 illustrates this observation and also shows

Table 3

Detailed cost values, total inventory levels, total number of facilities, shortage probabilities for $\kappa = 10$.

α	Base problem, risk coefficient $\lambda = 0$		Risk coefficient $\lambda = 0.1$		Risk coefficient $\lambda = 1$	
	0.7	0.8	0.7	0.8	0.7	0.8
Mean-risk func.	90.22×10^6	90.22×10^6	111.89×10^6	120.70×10^6	275.01×10^6	313.93×10^6
CVaR $_{\alpha}$	221.18×10^6	313.29×10^6	216.43×10^6	303.18×10^6	165.69×10^6	204.46×10^6
Total exp. cost	90.22×10^6	90.22×10^6	90.25×10^6	90.38×10^6	109.32×10^6	109.46×10^6
Positioning cost	28.81×10^6	28.81×10^6	30.50×10^6	31.04×10^6	70.71×10^6	70.89×10^6
Exp. recourse func.	61.42×10^6	61.42×10^6	59.75×10^6	59.34×10^6	38.61×10^6	38.57×10^6
Exp. trans. and salvage cost	4.76×10^6	4.76×10^6	5.11×10^6	5.20×10^6	13.01×10^6	13.03×10^6
Exp. shortage cost	56.66×10^6	56.66×10^6	54.64×10^6	54.14×10^6	25.60×10^6	25.54×10^6
Water (1000 gal)	8999.0	8999.0	8999.0	9020.7	17,735.7	17,667.0
Food (1000 meals)	3609.0	3609.0	3921.0	4001.0	9403.9	9403.9
Medical kits	18,749.0	18,749.0	18,753.4	19,109.0	47,111.7	48,732.0
Small facilities	6	6	6	8	6	6
Medium facilities	2	2	2	2	5	5
Large facilities	1	1	1	1	2	2
Shortage prob. water	0.093	0.093	0.093	0.093	0.205	0.205
Shortage prob. food	0.120	0.120	0.098	0.090	0.074	0.074
Shortage prob. med. kits	0.215	0.215	0.215	0.210	0.199	0.199

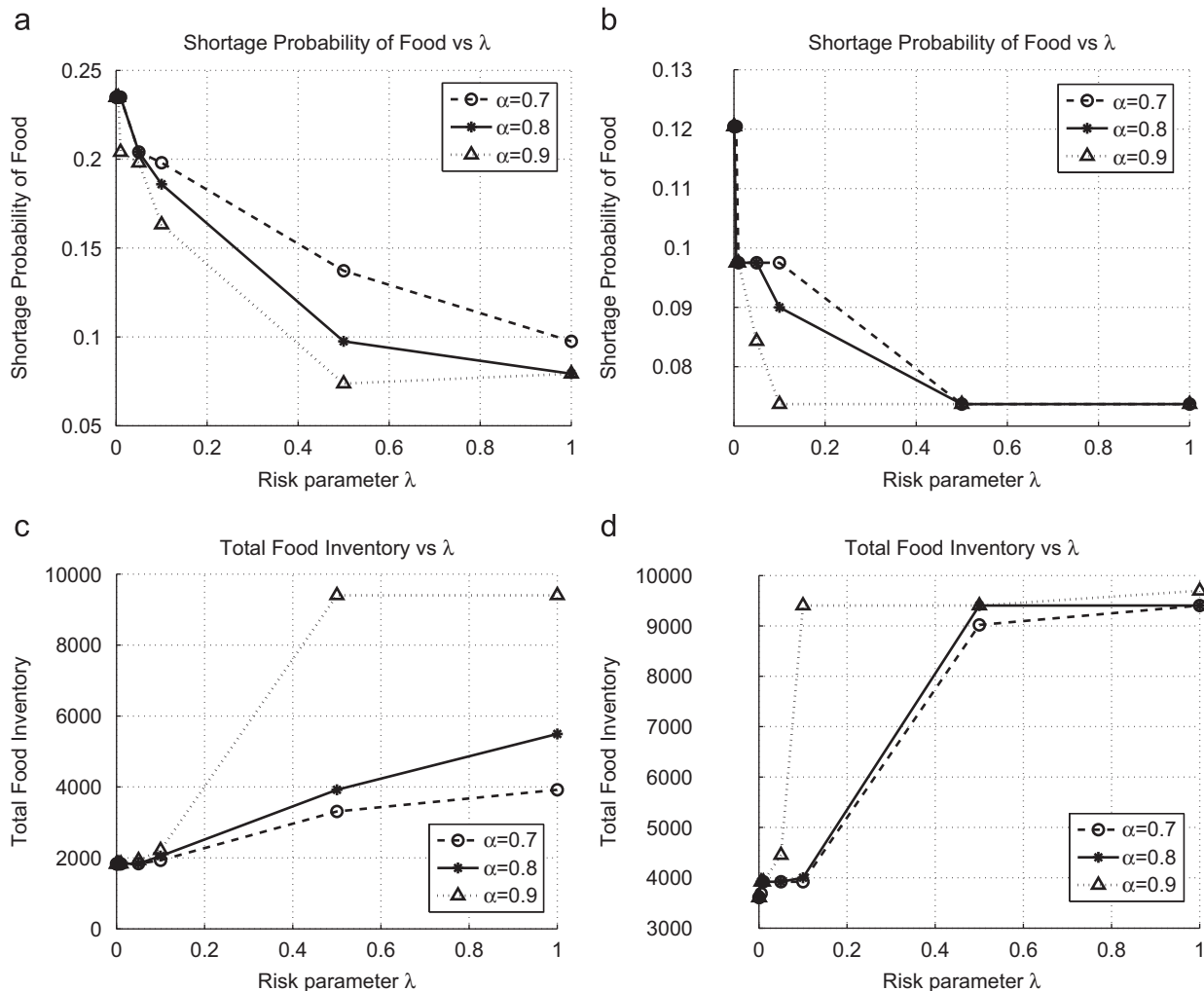


Fig. 3. Shortage probability and total inventory for food. (a,c) $\kappa = 5$. (b,d) $\kappa = 10$.

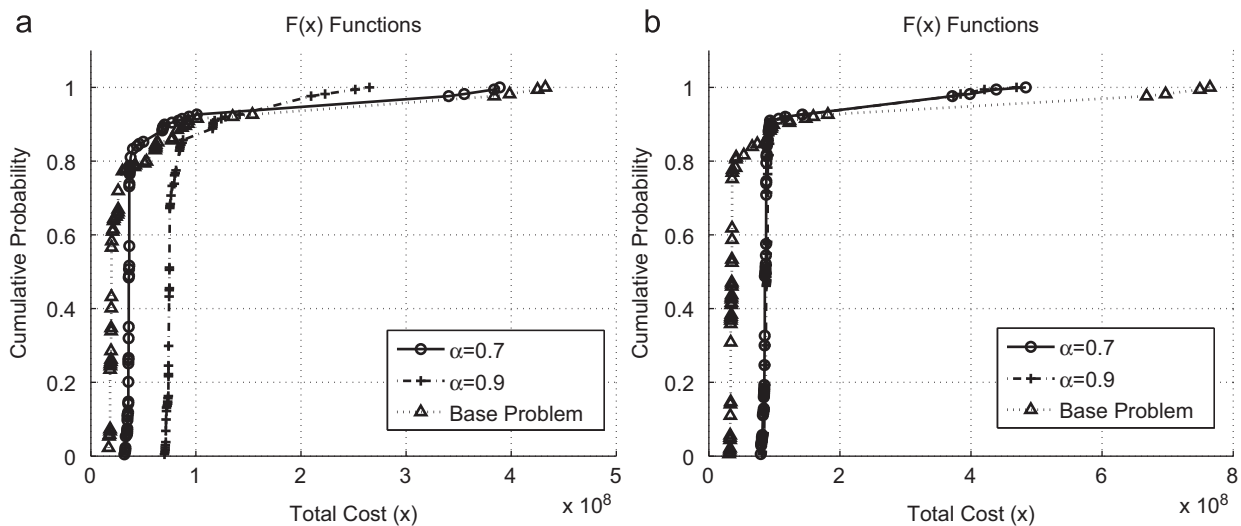


Fig. 4. Cumulative distribution functions of the total cost for different values of κ and λ parameters. (a) $\kappa = 5$ and $\lambda = 1$. (b) $\kappa = 10$ and $\lambda = 1$.

Table 4

Expected total cost and $\text{CVaR}_\alpha(f(\mathbf{r}, \mathbf{y}, \omega))$ for $\lambda = 0$.

κ	$\alpha = 0.7$		$\alpha = 0.8$		$\alpha = 0.9$	
	Expected total cost	CVaR_α	Expected total cost	CVaR_α	Expected total cost	CVaR_α
10	90.22×10^6	221.18×10^6	90.22×10^6	313.29×10^6	90.22×10^6	548.36×10^6
5	56.51×10^6	142.44×10^6	56.51×10^6	197.09×10^6	56.51×10^6	322.08×10^6

that the recourse cost and the total positioning cost, which together constitute the expected total cost, do not change monotonically as a function of α .

The total expected cost is decomposed into the positioning cost and the expected recourse cost; the expected recourse cost is decomposed into the expected transportation, salvage and shortage costs. These detailed cost values are presented in Tables 2 and 3. According to these results, increasing λ leads to a more risk-averse policy with higher positioning costs and lower expected recourse costs in general. Thus, a more risk-averse policy keeps higher inventory levels overall and results in lower expected shortage costs. This does not imply that the inventory level for each commodity increases; whether the inventory level of a commodity increases or decreases depends on the associated shortage cost. Recall that shortage costs for food and water are significantly higher with respect to the medical kits. Therefore, to avoid high shortage costs, the inventory levels of food and water increase significantly in contrast to the inventory level of medical kits for $\kappa = 5$ as in Table 2. However, Table 3 shows that for $\kappa = 10$ all shortage costs are high enough to lead an increase in the inventory level of each commodity.

Suppose that for a given first-stage variable we solve the second-stage problem and obtain the values of the shortage variables w_i^{ks} . If the second-stage variable w_i^{ks} is strictly positive for at least one node i , we say that the demand for commodity k is not fully satisfied under scenario s and there is a (demand) shortage. Basically, the shortage probability for commodity $k \in K$ measures the probability of violating the stochastic demand satisfaction constraints:

$$\sum_{(j,i) \in A} x_{ji}^k(\omega) - \sum_{(i,j) \in A} x_{ij}^k(\omega) + \gamma_i^k(\omega) r_i^k \geq \vartheta_i^k(\omega), \quad i \in I.$$

Table 5

Expected total cost versus $\text{CVaR}_\alpha(f(\mathbf{r}, \mathbf{y}, \omega))$ for $\kappa = 10$.

Risk coefficient, λ	$\alpha = 0.7$		$\alpha = 0.8$		$\alpha = 0.9$	
	Relative difference		Relative difference		Relative difference	
	Expected total cost (%)	CVaR_α (%)	Expected total cost (%)	CVaR_α (%)	Expected total cost (%)	CVaR_α (%)
0.005	0.006	−0.530	0.030	−2.612	0.030	−2.254
0.01	0.030	−2.150	0.030	−2.612	0.030	−2.254
0.05	0.030	−2.150	0.030	−2.612	1.120	−6.314
0.1	0.030	−2.150	0.173	−3.226	16.416	−40.419
0.5	15.988	−22.093	21.097	−34.658	18.723	−41.296
1	21.164	−25.089	21.322	−34.737	23.140	−42.169
5	21.916	−25.244	22.175	−34.895	26.907	−42.513
10	21.958	−25.247	22.183	−34.895	26.907	−42.513
50	21.991	−25.248	22.311	−34.897	29.075	−42.536
100	21.991	−25.248	22.399	−34.898	28.825	−42.537
1000	21.991	−25.248	22.404	−34.898	28.838	−42.537

Then for a given first-stage variable, the shortage probability for commodity k is calculated as

$$\sum_{s=1}^N \left\{ p^s : w_i^{ks} = \vartheta_i^k - \sum_{(j,i) \in A} x_{ji}^{ks} + \sum_{(i,j) \in A} x_{ij}^{ks} - \gamma_i^k r_i^k > 0 \text{ for at least one } i \in I \right\}.$$

In Tables 2 and 3 we report also the shortage probabilities for the (near-) optimal solutions. According to the results, a more risk-averse policy leads to higher inventory levels and lower shortage probabilities in general. When the shortage cost for commodity k is not significantly high, the model would provide a

solution for which the shortage of commodity k is more likely. This explains the higher shortage probability values for medical kits with respect to food and water. As evident from Table 3, more inventory is allocated for all commodities to obtain a lower expected shortage cost when κ increases. In this case, we also observe that the occurrence of shortage for each commodity is generally less likely. However, we cannot claim that the optimal solution of the proposed model would always perform well in terms of the shortage probabilities. The proposed model is concerned with the expected total shortage cost and not the shortage probability. For example, Table 3 presents an illustrative

example, which shows that a higher inventory level of water leads to a higher shortage probability. In addition, Fig. 3 shows how the shortage probability and the inventory level for food change with respect to the risk parameters.

In order to demonstrate the effect of incorporating the risk measure CVaR on the total cost, we obtain the empirical cumulative distribution of the random total cost associated with the (near-) optimal solution of the proposed model. When $\lambda = 0$, we have the risk-neutral two-stage stochastic programming formulation, which we refer to as the “Base Problem”. We solve the “Base Problem” and also the proposed risk-averse problem with CVaR for specified α values and obtain (near-) optimal policies. For a given policy, we can calculate the realization of the total cost under each scenario and derive the empirical cumulative distribution of the random total cost. Fig. 4 shows the cumulative distributions of the random total cost associated with the “Base Problem” and the proposed problem for $\alpha = 0.7$ and 0.9. Basically, the α parameter shapes the cumulative distribution according to the preferences of the decision makers. A larger α value avoids the occurrence of larger realization values, and therefore leads to a shift to the left in the right tail of the cumulative distribution function. As a trade-off, the expectation increases and it implies a shift to the right in the left tail of the cumulative distribution function.

As mentioned before, the proposed mean-risk model can support the decision process by constructing policies for different risk parameters; the policy makers can evaluate those policies according to their risk preferences. This is similar to constructing an efficient frontier in finance. In this spirit, we analyze the changes in the expected total cost versus the corresponding CVaR for different values of the risk parameters α and λ . Table 4 presents the expected total cost and CVaR of the total cost at a specified confidence level α for the optimal solution of the traditional expectation-based model, i.e., for $\lambda = 0$. Tables 5 and 6

Table 6Expected total cost versus $\text{CVaR}_\alpha(f(\mathbf{r}, \mathbf{y}, \omega))$ for $\kappa = 5$.

Risk coefficient, λ	$\alpha = 0.7$		$\alpha = 0.8$		$\alpha = 0.9$	
	Relative difference		Relative difference		Relative difference	
	Expected total cost (%)	CVaR $_\alpha$ (%)	Expected total cost (%)	CVaR $_\alpha$ (%)	Expected total cost (%)	CVaR $_\alpha$ (%)
0.005	0.000	0.000	0.000	0.000	0.000	−0.002
0.01	0.001	−0.036	0.000	0.000	0.003	−0.064
0.05	0.007	−0.140	0.007	−0.109	0.189	−0.733
0.1	0.331	−1.591	0.484	−1.858	0.989	−2.634
0.5	7.051	−10.013	10.326	−14.114	57.927	−38.871
1	10.710	−12.235	23.670	−18.893	58.046	−38.904
5	22.620	−13.729	69.792	−28.249	60.112	−39.058
10	26.826	−13.960	69.805	−28.249	63.048	−39.134
50	56.059	−14.517	69.821	−28.250	67.767	−39.186
100	56.059	−14.517	69.821	−28.250	67.767	−39.186
1000	56.062	−14.516	69.821	−28.250	67.781	−39.186

Table 7

Stochastic measures associated with the risk-neutral modeling approach.

	$\kappa = 5$		$\kappa = 10$			$\kappa = 5$		$\kappa = 10$	
RP	56.51	$\times 10^6$	90.23	$\times 10^6$	$\text{CVaR}_{0.7}(\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}, \omega))$	69.65	$\times 10^6$	108.15	$\times 10^6$
WS	25.97	$\times 10^6$	37.52	$\times 10^6$	$\text{CVaR}_{0.8}(\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}, \omega))$	96.13	$\times 10^6$	153.87	$\times 10^6$
EV	16.91	$\times 10^6$	16.91	$\times 10^6$	$\text{CVaR}_{0.9}(\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}, \omega))$	166.37	$\times 10^6$	280.34	$\times 10^6$
EEV	56.98	$\times 10^6$	95.11	$\times 10^6$	$\text{CVaR}_{0.7}(f(\bar{\mathbf{x}}(\xi), \omega))$	140.36	$\times 10^6$	262.96	$\times 10^6$
EVPI=RP−WS	30.54	$\times 10^6$	52.71	$\times 10^6$	$\text{CVaR}_{0.8}(f(\bar{\mathbf{x}}(\xi), \omega))$	193.48	$\times 10^6$	369.26	$\times 10^6$
EVPI/RP	54.05%		58.42%		$\text{CVaR}_{0.9}(f(\bar{\mathbf{x}}(\xi), \omega))$	316.36	$\times 10^6$	615.51	$\times 10^6$
VSS=EEV−RP	0.47	$\times 10^6$	4.88	$\times 10^6$					
VSS/RP	0.84%		5.41%						

Table 8

Stochastic measures associated with the mean-risk modeling approach.

α	Risk coefficient $\lambda = 0.5$			Risk coefficient $\lambda = 1$			Risk coefficient $\lambda = 5$		
	0.7	0.8	0.9	0.7	0.8	0.9	0.7	0.8	0.9
$\kappa = 5$									
MRRP	124.58×10^6	146.98×10^6	187.68×10^6	187.57×10^6	229.74×10^6	286.09×10^6	683.72×10^6	803.02×10^6	1071.89×10^6
MRVPI	63.79×10^6	72.95×10^6	78.53×10^6	91.96×10^6	107.64×10^6	93.75×10^6	309.50×10^6	296.41×10^6	214.05×10^6
MRVPI/MRRP	51.20%	49.63%	41.84%	49.02%	46.85%	32.77%	45.27%	36.91%	19.97%
MRVSS	2.58×10^6	6.74×10^6	27.48×10^6	9.77×10^6	20.72×10^6	87.25×10^6	75.07×10^6	221.35×10^6	566.87×10^6
MRVSS/MRRP	2.07%	4.59%	14.64%	5.21%	9.02%	30.50%	10.98%	27.56%	52.89%
$\kappa = 10$									
MRRP	190.81×10^6	211.61×10^6	268.07×10^6	275.01×10^6	313.93×10^6	428.23×10^6	936.74×10^6	1130.07×10^6	1690.70×10^6
MRVPI	99.22×10^6	97.16×10^6	90.39×10^6	129.35×10^6	122.54×10^6	110.37×10^6	358.50×10^6	323.20×10^6	251.49×10^6
MRVPI/MRRP	52.00%	45.92%	33.72%	47.03%	39.03%	25.77%	38.27%	28.60%	14.87%
MRVSS	35.78×10^6	68.12×10^6	134.79×10^6	83.06×10^6	150.44×10^6	282.39×10^6	473.18×10^6	811.32×10^6	1481.98×10^6
MRVSS/MRRP	18.75%	32.19%	50.28%	30.20%	47.92%	65.95%	50.51%	71.79%	87.65%

report the relative differences in the (near-) optimal expectation and CVaR values of the total cost obtained by the risk-averse model with respect to the corresponding values obtained by the risk-neutral model.

Here we focus on computing the stochastic measures, which are designed to indicate whether it is worth to model the randomness by using a risk-neutral or a risk-averse stochastic programming approach. For the risk-neutral approach, we compute the EVPI and the VSS and provide the results in Table 7. For $\kappa = 5$ the VSS is significantly small relative to the RP value, which indicates that using the solution of the expected value problem is likely to result in an objective function value which is really close to the optimal objective function value of the risk-neutral two-stochastic programming problem. Thus, for this specific case study when the penalty cost of unmet demand is not significantly high, modeling the randomness based on the risk-neutral approach does not play a significant role on the optimal solutions. When the penalty cost increases ($\kappa = 10$), the VSS is higher and then it may be worth to solve the risk-neutral stochastic problem if the preference relation is only based on the expected values. However, as illustrated above, the risk-neutral approach may not be sufficient to avoid larger realizations of the total cost values. Therefore, we may prefer to use a risk-averse stochastic model. Then to check the significance of solving the risk-averse model instead of solving the simpler deterministic versions of the problem, we compute the measures defined in Section 4 for the risk-averse approach. Table 7 also provides the CVaR values associated with the risk-neutral WS and EV approaches, which are used to compute our proposed measures. Table 8 presents the results on the measures for the proposed risk-averse stochastic model. As seen from these results, MRVSS values are in general significantly high and the value of MRVSS increases when we increase the level of risk aversion, i.e., when we increase the values of the risk parameters λ and α . These results indicate that solving the risk-averse model is significant to obtain solutions which are preferred with respect to the specified risk preferences. According to the numerical results, MRVPI values are also high, but they get smaller when we increase the level of risk aversion.

In Table 9, we present the CPU times and the upper bounds on the relative optimality gaps of the proposed algorithms for the case study instances. “CPU Direct” refers to the CPU time required to solve the direct formulation of DALPWithCVaR (DEF) by CPLEX. Solving DEF by CPLEX provides us with an optimal objective function value f^* . Let $\hat{f}$ denote the objective function value of the solution found by a decomposition algorithm when it stops within tolerance ε ($\varepsilon = 0.015$). Then, we calculate the upper bound on the relative optimality gap associated with the objective

function value $\hat{f}$ as follows:

$$\text{UBROG}^* = \frac{\hat{f} - f^*}{f^*}.$$

As seen from Table 9, solving the case study instances using the direct formulation might take less CPU time, especially for large λ values. The direct approach might outperform a decomposition algorithm for a small number of scenarios, but it is well-known that a decomposition algorithm would be computationally more efficient for a larger set of scenarios. In order to show some numerical results on the computational effectiveness of the proposed algorithms, we randomly generate problem instances, as discussed in the following section, using parameters similar to those in the case study.

6.2. Computational performance of the proposed algorithms

We generated random problem instances of different sizes to illustrate the computational performance of the proposed algorithms. We randomly generated 50 nodes on the plane and then constructed links between the nodes according to a specified threshold distance in order to obtain a disaster network consisting of 50 nodes. We generated the demand value at each node for a particular commodity under each scenario as follows: First, we

Table 10
Dimensions of the problem instances.

Instance number	Scenarios	Links	Continuous vars.	Constraints
1	$N=100$	594	208,450	74,551
2		636	221,050	78,751
3		492	177,850	64,351
4		596	2090,50	74,751
5		534	190,450	68,551
6	$N=200$	430	318,350	116,251
7		600	420,350	150,251
8		464	338,750	123,051
9		630	438,350	156,251
10		438	323,150	117,851
11	$N=300$	422	470,250	171,951
12		586	617,850	221,151
13		434	481,050	175,551
14		712	731,250	258,951
15		444	490,050	178,551
16	$N=500$	422	1,059,650	378,551
17		606	783,650	286,551
18		406	759,650	278,551
19		624	1,086,650	387,551
20		458	837,650	304,551

Table 9
CPU times and upper bounds on the relative optimality gap.

	CPU Times			# of Iterations		UBROG*	
	Direct	Algorithm 1	Algorithm 2	Algorithm 1	Algorithm 2	Algorithm 1 (%)	Algorithm 2 (%)
$\lambda = 0.1$							
$\alpha = 0.7$	269.32	300.07	86.23	40	52	0.44	0.52
$\alpha = 0.8$	533.25	288.45	63.84	40	42	0.49	0.41
$\alpha = 0.9$	503.68	135.51	72.46	33	46	0.43	0.31
$\lambda = 0.5$							
$\alpha = 0.7$	162.81	311.37	128.32	29	71	0.29	0.38
$\alpha = 0.8$	117.86	160.75	103.73	31	65	0.36	0.49
$\alpha = 0.9$	102.64	405.34	132.67	45	74	0.64	0.78
$\lambda = 1$							
$\alpha = 0.7$	201.07	266.47	256.37	23	102	0.33	0.63
$\alpha = 0.8$	101.19	557.64	328.98	30	115	0.39	0.39
$\alpha = 0.9$	70.72	544.46	173.93	49	83	0.52	0.89

randomly selected a node as the center of the disaster. Then, for a random number of nodes in the neighborhood we randomly generated demand values based on the distances from the center node; the generated demand for a node that is further away from the center node is smaller than the demand at a node that is closer to the center of the disaster. Table 10 presents the dimensional properties of the generated test problem instances.

First, we would like to emphasize that solving DEF directly using a standard mixed integer programming solver such as CPLEX is hard for large problem instances. Thus, most of the generated problem instances cannot be solved for optimality within the prescribed time limit and therefore, in order to calculate the relative optimality gap we use the best known lower bound on the objective value found by CPLEX. Let Obf_t denote the best lower bound on the objective function value that is provided by the CPLEX solver, when the prescribed time limit t is reached ($t=7200$ s). Obf_t^* denotes the best available objective function value within the time limit, which defines an upper bound on the objective value. Then, we define an upper bound on

the relative optimality gap as follows:

$$\text{UBROG}_t = \frac{\text{Obf}_t^* - \text{Obf}_t}{\text{Obf}_t}.$$

We also calculate the upper bound on the relative optimality gap for the best objective function value obtained by the proposed algorithms for the stopping tolerance $\varepsilon = 0.015$:

$$\text{UBROG}_t = \frac{\hat{f} - \text{Obf}_t}{\text{Obf}_t}.$$

To have a fair comparison, we also set the mixed-integer optimality gap tolerance parameter (mipgap) to 0.015 for the CPLEX solver. As noted in Table 11, CPLEX could not provide a lower bound on the optimal objective function value for some problem instances. In this case, the maximum of the objective function values obtained by the master problems at termination of Algorithms 1 and 2 is taken as the best available lower bound. According to Table 11, the proposed single-cut decomposition algorithms provide solutions with very small optimality gaps in

Table 11

CPU times, the number of optimality cuts, and the upper bounds on the relative optimality gap for $\lambda = 1$.

Instance number	CPU: (# of Iterations)		CPU direct	UBROG _t			Relative reduction in CPU	
	Algorithm 1	Algorithm 2		Alg. 1 (%)	Alg. 2 (%)	Direct (%)	Algorithm 1 (%)	Algorithm 2 (%)
$\alpha = 0.9$								
1	164.49; (13)	234.31; (36)	1635.71	0.56	0.80	0.62	89.94	85.68
2	119.16; (11)	537.32; (55)	1917.13	0.48	1.12	0.45	93.78	71.97
3	106.92; (11)	363.64; (50)	898.09	0.67	0.94	1.00	88.09	59.51
4	151.35; (11)	308.04; (43)	1378.11	0.45	0.64	0.45	89.02	77.65
5	419.89; (14)	527.27; (56)	347.27	1.07	0.98	1.44	−20.91	−51.83
6	508.73; (12)	468.39; (41)	3083.51	0.48	0.60	0.58	83.50	84.81
7	161.97; (10)	657.55; (46)	4965.30	0.80	0.67	0.52	96.74	86.76
8	905.83; (16)	553.20; (46)	7205.62	0.74	0.74	1.71	> 87.43	> 92.32
9	517.74; (12)	1094.68; (59)	5156.86	0.43	1.18	0.42	89.96	78.77
10	454.95; (12)	301.49; (32)	6338.66	0.71	0.56	0.61	92.82	95.24
11	498.36; (11)	637.88; (40)	6123.73	0.84	0.54	0.50	91.86	89.58
12	707.84; (12)	1421.39; (57)	7219.68	1.25 ^a	1.23 ^a	100.00	> 90.20	> 80.31
13	1089.23; (14)	770.07; (46)	7209.08	0.79	0.76	1.72	> 84.89	> 89.32
14	655.73; (11)	947.10; (42)	6434.75	0.43	0.60	0.40	89.81	85.28
15	918.35; (13)	486.69; (34)	5845.66	0.66	0.79	0.43	84.29	91.67
16	2684.76; (13)	1273.57; (46)	7251.32	0.84	0.94	10.93	> 62.98	> 82.44
17	1114.60; (13)	3567.46; (73)	7241.72	1.06 ^a	1.38 ^a	100.00	> 84.61	> 50.74
18	3539.52; (16)	1268.51; (46)	7233.41	1.05 ^a	1.05 ^a	100.00	> 51.07	> 82.46
19	1426.73; (11)	1535.98; (45)	7255.26	1.08 ^a	1.28 ^a	100.00	> 80.34	> 78.83
20	1723.20; (12)	1050.51; (40)	7243.75	1.41 ^a	1.17 ^a	100.00	> 76.21	> 85.50
$\alpha = 0.7$								
1	264.74; (14)	194.33; (32)	1422.08	0.48	0.65	0.42	81.38	86.33
2	186.66; (13)	180.88; (30)	1831.26	0.48	0.58	0.42	89.81	90.12
3	302.88; (11)	306.69; (44)	1116.67	0.62	0.92	1.04	72.88	72.54
4	129.19; (10)	255.05; (35)	1829.69	0.67	0.92	0.45	92.94	86.06
5	548.52; (16)	306.73; (42)	566.27	1.15	1.16	1.47	3.13	45.83
6	1581.94; (15)	352.32; (33)	5239.41	0.53	0.68	0.53	69.81	93.28
7	678.18; (13)	684.41; (44)	6630.70	0.50	1.09	0.51	89.77	89.68
8	1092.36; (16)	514.95; (43)	5843.85	0.64	0.81	1.07	81.31	91.19
9	1007.16; (12)	465.90; (37)	6820.72	0.61	0.56	0.48	85.23	93.17
10	493.28; (11)	292.05; (31)	5894.88	0.59	0.81	0.44	91.63	95.05
11	837.19; (13)	553.43; (37)	7218.93	1.08 ^a	1.33 ^a	100.00	> 88.40	> 92.33
12	1099.49; (11)	633.11; (35)	7221.57	1.08 ^a	1.44 ^a	100.00	> 84.77	> 91.23
13	2927.85; (15)	670.90; (41)	7219.78	1.29 ^a	1.21 ^a	100.00	> 59.45	> 90.71
14	862.53; (11)	541.13; (30)	7219.99	1.04 ^a	1.24 ^a	100.00	> 88.05	> 92.51
15	1787.62; (12)	334.46; (27)	7219.53	0.53	0.65	11.45	> 75.24	> 95.37
16	3799.04; (14)	1192.49; (42)	7240.30	0.93 ^a	1.23 ^a	100.00	> 47.53	> 83.53
17	3042.06; (11)	1476.19; (44)	7246.99	0.94 ^a	1.29 ^a	100.00	> 58.02	> 79.63
18	3607.01; (14)	1036.28; (40)	7229.07	1.28 ^a	1.18 ^a	100.00	> 50.10	> 85.67
19	1542.06; (12)	787.91; (30)	7246.43	1.27 ^a	1.22 ^a	100.00	> 78.72	> 89.13
20	2392.59; (12)	729.14; (32)	7235.35	1.23 ^a	1.20 ^a	100.00	> 66.93	> 89.92

The CPU times exceeding the time limit are in boldface.

^a CPLEX could not provide a lower bound on the optimal objective function value.

Table 12Multicut versions: CPU times, the number of optimality cuts, and the upper bounds on the relative optimality gap for $\lambda = 1$.

Instance number	Reduction in CPU wrt to the single-cut version Algorithm 2 with multicuts			Reduction in # of Iterations wrt to the single-cut version Algorithm 2 with multicuts			Reduction in UBROG _r wrt to the single-cut version Algorithm 2 with multicuts		
	(31) (%)	(29) and (30) (%)	(30) (%)	(31) (%)	(29) and (30) (%)	(30) (%)	(31) (%)	(29) and (30) (%)	(30) (%)
$\alpha = 0.9$									
1	51.71	67.32	51.53	69.44	72.22	61.11	4.00	−0.17	0.21
2	55.13	62.35	226.50	74.55	80.00	76.36	4.21	0.57	0.57
3	71.52	41.37	186.94	78.00	82.00	76.00	3.66	0.04	0.03
4	55.26	70.34	462.74	74.42	76.74	74.42	5.95	0.07	−0.02
5	56.86	74.42	185.82	76.79	82.14	75.00	7.47	−0.39	−0.11
6	−74.59	−22.16	77.31	70.73	73.17	70.73	3.91	0.07	−0.03
7	46.54	69.73	183.36	73.91	78.26	76.09	4.37	0.07	0.13
8	21.86	−4.14	50.86	76.09	76.09	67.39	4.75	−0.44	0.11
9	60.01	77.59	456.21	79.66	83.05	81.36	4.13	0.71	0.67
10	−102.85	11.41	71.92	62.50	68.75	65.63	5.81	0.06	−0.23
11	19.40	44.44	238.06	72.50	75.00	72.50	5.70	0.00	−0.02
12	59.12	71.15	575.67	80.70	82.46	82.46	3.40	0.19	−0.07
13	−8.95	−50.90	47.62	73.91	78.26	71.74	5.11	0.03	−0.28
14	43.79	69.07	510.69	71.43	76.19	76.19	4.60	−0.40	0.05
15	−331.03	−387.61	− 34.60	61.76	61.76	58.82	5.99	0.30	0.28
16	−100.91	71.88	227.23	73.91	80.43	76.09	6.22	−0.13	0.02
17	67.71	−25.21	468.50	83.56	83.56	83.56	4.95	0.40	0.36
18	−110.65	24.99	36.80	71.74	80.43	76.09	5.88	−0.09	−0.07
19	53.94	−86.64	648.86	75.56	75.56	80.00	4.80	0.20	0.08
20	−35.55	−162.10	1.76	72.50	72.50	67.50	5.09	−0.53	−0.15
$\alpha = 0.7$									
1	55.66	79.65	43.18	65.63	75.00	62.50	6.38	−0.37	0.07
2	−50.05	3.10	54.21	53.33	63.33	63.33	4.29	0.13	0.02
3	47.27	4.62	58.68	72.73	75.00	75.00	3.82	−0.16	0.15
4	45.00	45.15	75.78	68.57	71.43	71.43	6.58	0.38	0.34
5	10.79	−407.20	−64.02	71.43	76.19	73.81	5.90	0.04	−0.03
6	−119.92	−323.26	− 50.92	63.64	63.64	60.61	5.66	0.22	0.18
7	19.76	60.94	76.37	72.73	77.27	77.27	5.23	0.53	0.41
8	−17.21	−122.27	1.84	69.77	69.77	67.44	4.94	−0.12	0.19
9	−19.89	−24.89	9.23	70.27	72.97	67.57	2.54	0.10	0.11
10	−217.50	−727.33	− 7.85	58.06	58.06	61.29	6.23	0.35	0.28
11	19.47	6.01	69.95	67.57	72.97	72.97	5.85	0.34	0.23
12	40.46	−7.94	18.41	71.43	71.43	71.43	3.87	0.39	0.34
13	−11.31	−347.82	13.89	73.17	70.73	70.73	4.89	0.02	−0.05
14	7.42	−391.29	−39.45	63.33	63.33	60.00	3.52	0.12	0.02
15	− 309.06	−309.99	−366.87	51.85	59.26	44.44	5.96	−0.44	0.17
16	−133.51	−150.37	− 94.16	71.43	73.81	69.05	6.72	0.44	0.44
17	−135.17	−162.89	21.51	72.73	77.27	72.73	4.97	0.38	0.41
18	−68.34	− 5.87	−39.81	70.00	77.50	75.00	6.62	0.16	0.10
19	−95.52	−376.75	46.14	63.33	66.67	66.67	4.52	0.06	0.06
20	−599.69	−422.53	− 84.12	56.25	65.63	65.63	6.60	0.17	−0.06

The best reduction percentages are in boldface.

reasonable CPU times. We can also state that the single-cut subgradient-based decomposition algorithm requires the generation of significantly more optimality cuts.

Since the number of the iterations for the single-cut version of Algorithm 1 is already small, we decided to report results for the multicut versions of Algorithm 2 in order to demonstrate how the multicut approach affects the computational performance. We consider three different multicut versions of Algorithm 2; the first, the second and the third one involves the optimality cuts (31), (29) and (30), and (30), respectively. In Table 12, we report the CPU times, the number of iterations and the upper bound values on the relative optimality gaps with respect to the results of Algorithm 2 presented in Table 11. According to Table 12, using multicuts (30) significantly reduces the computational times for $\alpha = 0.9$. However, the results for $\alpha = 0.7$ show that the multicut versions of the decomposition algorithms may be computationally expensive. As expected according to the discussions in Section 3.3, Table 12 illustrates that the multicut methods significantly reduce the number of iterations, but the trade-off

between the number of iterations and the size of the master problem is instance-dependent.

7. Conclusion and future work

In this study, we consider a two-stage mean-risk stochastic programming model, which incorporates the trade-off between the expected total cost and a risk measure on the random total cost. Among the downside risk measures we focus on a recently popular and widely applied risk measure CVaR. We also adapt the concepts of the EVPI and VSS for the proposed risk-averse two-stage stochastic programming models and define two measures on the VPI and VSS. The main contributions of our study are developing solution algorithms for the proposed model with CVaR and introducing new stochastic measures on the VPI and VSS. As an application, we consider a disaster management problem that can significantly benefit from the risk-averse two-stage stochastic programming models. The proposed problem can be solved for

different risk parameters, and this would allow the decision makers to evaluate different location and allocation decisions based on the trade-off between the quality of service and costs. The computational study illustrates how the risk parameters affect the location and allocation policies and shows that we can solve the proposed model for a reasonably large number of scenarios. Then, our additional contributions are listed as developing a new risk-averse location and allocation model for a disaster relief management under uncertainty in demand and the damage level of the disaster network, and showing that the proposed risk-averse disaster management model can be solved efficiently for reasonably large problem instances. In our future research, we would like to also consider decisions on transportation capacities and develop new stochastic disaster relief network design models.

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References

- [1] Prékopa A. Stochastic programming. Dordrecht, Boston: Kluwer Academic; 1995.
- [2] Birge J, Louveaux F. Introduction to stochastic programming. New York: Springer; 1997.
- [3] Ahmed S. Mean-risk objectives in stochastic programming. Technical Report, Georgia Institute of Technology. E-print available at 2004, <<http://www.optimization-online.org>>.
- [4] Ahmed S. Convexity and decomposition of mean-risk stochastic programs. Mathematical Programming 2006;106(3):433–46.
- [5] Schultz R, Tiedemann S. Conditional value-at-risk in stochastic programs with mixed-integer recourse. Mathematical Programming 2006;105(2):365–86.
- [6] Fábian CI. Handling CVAR objectives and constraints in two-stage stochastic models. European Journal of Operational Research 2008;191(3):888–911.
- [7] Miller N, Ruszczyński A. Risk-averse two-stage stochastic linear programming: modeling and decomposition. Optimization Online Digest 2009; August.
- [8] Küenzi-Bay A, Mayer J. Computational aspects of minimizing conditional value-at-risk. Computational Management Science 2006;3:3–27.
- [9] Klein Haneveld WK, Van der Vlerk MH. Integrated chance constraints: reduced forms and an algorithm. Computational Management Science 2006;3:245–69.
- [10] Van Slyke RM, Wets R. L-shaped linear programs with applications to optimal control and stochastic programming. SIAM Journal on Applied Mathematics 1969;17(4):638–63.
- [11] Benders JF. Partitioning procedures for solving mixed-variable programming problems. Numerische Mathematik 1962;54:238–52.
- [12] Schultz R, Neise F. Algorithms for mean-risk stochastic integer programs in energy. Revista Investigación Operacional 2007;28:4–16.
- [13] Liu C, Fan Y, Ordóñez F. A two-stage stochastic programming model for transportation network protection. Computers and Operations Research 2009;36:1582–90.
- [14] Barbarosoğlu G, Arda Y. A two-stage stochastic programming framework for transportation planning in disaster response. Journal of the Operational Research Society 2004;55:43–53.
- [15] Balci B, Beamon B. Facility location in humanitarian relief. International Journal of Logistics: Research and Applications 2008;11(2):101–21.
- [16] Rawls CG, Turnquist MA. Pre-positioning of emergency supplies for disaster response. Transportation Research Part B: Methodological 2010;44(4):521–34.
- [17] Rockafellar R, Uryasev S. Conditional value-at-risk for general loss distributions. Journal of Banking and Finance 2002;26:1443–71.
- [18] Chen G, Daskin M, Shen M, Uryasev S. The alpha-reliable mean-excess regret model for stochastic facility location modeling. Naval Research Logistics 2006;53(7):617–26.
- [19] Markowitz MH. Portfolio selection. Journal of Finance 1952;7(1):77–91.
- [20] Ogryczak W, Ruszczyński A. Dual stochastic dominance and related mean-risk models. SIAM Journal of Optimization 2002;13(1):60–78.
- [21] Rockafellar R, Uryasev S. Optimization of conditional value at risk. The Journal of Risk 2000;2(3):21–41.
- [22] Pflug G. Some remarks on the value-at-risk and the conditional value-at-risk. In: Uryasev S, editor. Probabilistic constrained optimization: methodology and applications. Dordrecht: Kluwer Academic Publishers; 2000.
- [23] Birbil SI, Frenk J, Kaynar B, Noyan N. Risk measures and their applications in asset management. In: Gregoriou GN, editor. The VaR implementation handbook. New York: The McGraw-Hill Companies; 2009. p. 311–37.
- [24] Shaked M, Shanthikumar JG. Stochastic orders and their applications. Boston: Associated Press; 1994.
- [25] Müller A, Stoyan D. Comparison methods for stochastic models and risks. Chichester: John Wiley & Sons; 2002.
- [26] Ogryczak W, Ruszczyński A. From stochastic dominance to mean-risk models: semideviations as risk measures. European Journal of Operational Research 1999;116:33–50.
- [27] Laporte G, Louveaux F. The integer L-shaped method for stochastic integer programs with complete recourse. Operations Research Letters 1993;13(3):133–42.
- [28] Birge JR, Louveaux FV. A multicut algorithm for two-stage stochastic linear programs. European Journal of Operational Research 1988;34:384–92.
- [29] Gassmann HI. MSLiP: a computer code for the multistage stochastic linear programming problem. Mathematical Programming 1990;47:407–23.
- [30] Ruszczyński A. A regularized decomposition method for minimizing a sum of polyhedral functions. Mathematical Programming 1986;35:309–33.
- [31] Ruszczyński A, Świetanowski A. On the regularized decomposition method for two stage stochastic linear problems. Technical Report, International Institute for Applied Systems Analysis, WP-96-014; 1996.
- [32] Ruszczyński A. Decomposition methods. In: Ruszczyński A, Shapiro A, editors. Handbooks in operations research and management science, vol. 10. Amsterdam: Elsevier; 2003.
- [33] Kall P, Mayer J. Stochastic linear programming: models, theory, and computation. International series in operations research and management science. Springer; 2005.
- [34] Artzner P, Delbaen F, Eber JM, Heath D. Coherent measures of risk. Mathematical Finance 1999;9:203–27.
- [35] Ruszczyński A, Shapiro A. Optimization of risk measures. In: Calafiore G, Dabbene F, editors. Probabilistic and randomized methods for design under uncertainty. London: Springer-Verlag; 2006.
- [36] Fourer R, Gay D, Kernighan B. AMPL: a modeling language for mathematical programming, 2nd ed. Duxbury Press Brooks Cole Publishing Co; 2003.
- [37] ILOG, ILOG AMPL CPLEX System Version 11 User's Guide. ILOG CPLEX Division; 2008.